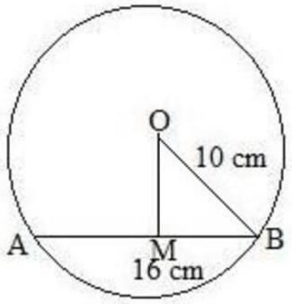
Exercise 12A

Solution 1: Let AB be the chord of the given circle with centre O and a radius of 10 cm.

Then AB = 16 cm and OB = 10 cm



From O, draw OM perpendicular to AB.

We know that the perpendicular from the centre of a circle to a chord bisects the chord.

$\therefore$ BM $= \left(\frac{16}{2}\right)$ cm $= 8$ cm

In the right ΔOMB, we have:

$OB^2 = OM^2 + MB^2$ (Pythagoras theorem)

$\Rightarrow 10^2 = OM^2 + 8^2$

$\Rightarrow 100 = OM^2 + 64$

$\Rightarrow OM^2 = (100 - 64)$

$\Rightarrow OM^2 = 36$

$\Rightarrow OM = \sqrt{36}$ cm

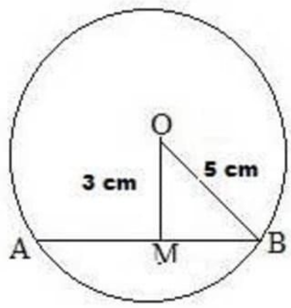
$\Rightarrow OM = 6$ cm

Hence, the distance of the chord from the centre is 6 cm.

Solution 2: Let AB be the chord of the given circle with centre O and a radius of 5 cm.

From O, draw OM perpendicular to AB.

Then OM = 3 cm and OB = 5 cm



From the right ΔOMB, we have:

$OB^2 = OM^2 + MB^2$ (Pythagoras theorem)

$\Rightarrow 5^2 = 3^2 + MB^2$

$\Rightarrow 25 = 9 + MB^2$

$\Rightarrow MB^2 = (25 - 9)$

$\Rightarrow MB^2 = 16$

$\Rightarrow MB = \sqrt{16}$ cm

$\Rightarrow MB = 4$ cm

Since the perpendicular from the centre of a circle to a chord bisects the chord, we have:

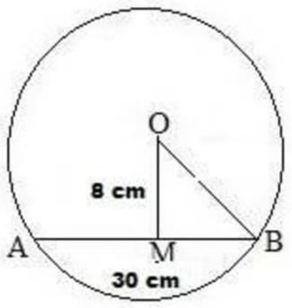
$AB = 2 \times MB = (2 \times 4)$ cm $= 8$ cm

Hence, the required length of the chord is 8 cm.

Solution 3: Let AB be the chord of the given circle with centre O. The perpendicular distance from the centre of the circle to the chord is 8 cm.

Join OB.

Then OM = 8 cm and AB = 30 cm



We know that the perpendicular from the centre of a circle to a chord bisects the chord.

$\therefore MB = \left(\frac{AB}{2}\right) = \left(\frac{30}{2}\right)$ cm $= 15$ cm

From the right ΔOMB, we have:

$OB^2 = OM^2 + MB^2$

$\Rightarrow OB^2 = 8^2 + 15^2$

$\Rightarrow OB^2 = 64 + 225$

$\Rightarrow OB^2 = 289$

$\Rightarrow OB = \sqrt{289}$ cm

$\Rightarrow OB = 17$ cm

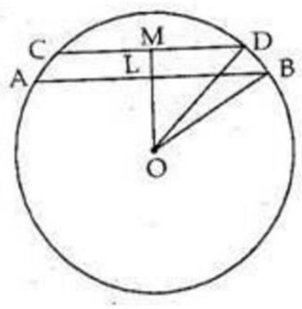
Hence, the required length of the radius is 17 cm.

Solution 4: We have:

(i) Let AB and CD be two chords of a circle such that AB is parallel to CD on the same side of the circle.

Given: AB = 8 cm, CD = 6 cm and OB = OD = 5 cm

Join OL and OM.



The perpendicular from the centre of a circle to a chord bisects the chord.

$\therefore LB = \frac{AB}{2} = \left(\frac{8}{2}\right) = 4cm$

Now, in right angled ΔBLO, we have:

$OB^2 = LB^2 + LO^2$

$\Rightarrow LO^2 = OB^2 - LB^2$

$\Rightarrow LO^2 = 5^2 - 4^2$

$\Rightarrow LO^2 = 25 - 16$

$\Rightarrow LO^2 = 9$

$\Rightarrow LO = 3$ cm

Similarly, $MD = \frac{CD}{2} = \frac{6}{2} = 3$ cm

In right angled ΔDMO, we have:

$OD^2 = MD^2 + MO^2$

$\Rightarrow MO^2 = OD^2 - MD^2$

$\Rightarrow MO^2 = 5^2 - 3^2$

$\Rightarrow MO^2 = 25 - 9$

$\Rightarrow MO^2 = 16$

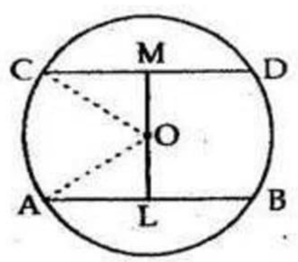
$\Rightarrow MO = 4$ cm

$\therefore$ Distance between the chords $= (MO - LO) = (4 - 3)$ cm $= 1$ cm

(ii) Let AB and CD be two chords of a circle such that AB is parallel to CD and they are on the opposite sides of the centre.

Given: AB = 8 cm and CD = 6 cm

Draw OL ⊥ AB and OM ⊥ CD.



Join OA and OC.

OA = OC = 5 cm (Radii of a circle)

The perpendicular from the centre of a circle to a chord bisects the chord.

$\therefore \mathrm{AL} = \frac{\mathrm{AB}}{2} = \frac{8}{2} = 4\mathrm{cm}$

Now, in right angled ΔOLA, we have:

$\mathrm{OA}^2 = \mathrm{AL}^2 + \mathrm{LO}^2$

$\Rightarrow \mathrm{LO}^2 = \mathrm{OA}^2 - \mathrm{AL}^2$

$\Rightarrow \mathrm{LO}^2 = 5^2 - 4^2$

$\Rightarrow \mathrm{LO}^2 = 25 - 16 = 9$

$\therefore \mathrm{LO} = 3$ cm

Similarly, $\mathrm{CM} = \frac{\mathrm{CD}}{2} = \frac{6}{2} = 3\mathrm{cm}$

In right angled ΔCMO, we have:

$\mathrm{OC}^2 = \mathrm{CM}^2 + \mathrm{MO}^2$

$\Rightarrow \mathrm{MO}^2 = \mathrm{OC}^2 - \mathrm{CM}^2$

$\Rightarrow \mathrm{MO}^2 = 5^2 - 3^2$

$\Rightarrow \mathrm{MO}^2 = 25 - 9$

$\Rightarrow \mathrm{MO}^2 = 16$

$\Rightarrow \mathrm{MO} = 4$ cm

Hence, distance between the chords = (MO + LO)

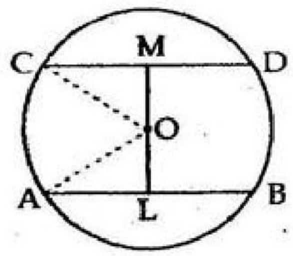
= (4 + 3) cm

$= 7$ cm

Solution 5: Let AB and CD be two chords of a circle such that AB is parallel to CD and they are on the opposite sides of the centre.

Given: AB $= 30$ cm and CD $= 16$ cm

Draw OL $\perp$ AB and OM $\perp$ CD.



Join OA and OC.

OA $=$ OC $= 17$ cm (Radii of a circle)

The perpendicular from the centre of a circle to a chord bisects the chord.

$\therefore \text{AL} = \frac{\text{AB}}{2} = \frac{30}{2} = 15 \text{ cm}$

Now, in right angled ΔOLA, we have:

$\text{OA}^2 = \text{AL}^2 + \text{LO}^2$

$\Rightarrow \text{LO}^2 = \text{OA}^2 - \text{AL}^2$

$\Rightarrow \text{LO}^2 = 17^2 - 15^2$

$\Rightarrow \text{LO}^2 = 289 - 225$

$\Rightarrow \text{LO}^2 = 64$

$\therefore \text{LO} = 8$ cm

Similarly, $\text{CM} = \left(\frac{\text{CD}}{2}\right) = \frac{16}{2} = 8 \text{ cm}$

In right angled ΔCMO, we have:

$\Rightarrow \text{OC}^2 = \text{CM}^2 + \text{MO}^2$

$\Rightarrow \text{MO}^2 = \text{OC}^2 - \text{CM}^2$

$\Rightarrow MO^2 = 17^2 - 8^2$

$\Rightarrow MO^2 = 289 - 64$

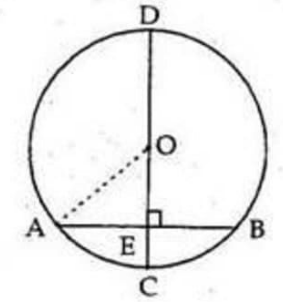
$\Rightarrow MO^2 = 225$

$\therefore MO = 15$ cm

Hence, distance between the chords = (LO + MO) = (8 + 15) cm = 23 cm

Solution 6: CD is the diameter of the circle with centre O and is perpendicular to chord AB.

Join OA.



Given: AB = 12 cm and CE = 3 cm

Let OA = OC = r cm (Radii of a circle)

Then OE = (r − 3) cm

Since the perpendicular from the centre of the circle to a chord bisects the chord, we have:

$$AE = \frac{AB}{2} = \frac{12}{2} \text{ cm} = 6 \text{ cm}$$

Now, in right angled ΔOEA, we have:

$\Rightarrow OA^2 = OE^2 + AE^2$

$\Rightarrow r^2 = (r - 3)^2 + 6^2$

$\Rightarrow r^2 = r^2 - 6r + 9 + 36$

$\Rightarrow r^2 - r^2 + 6r = 45$

$\Rightarrow 6r = 45$

$\Rightarrow r = \dfrac{45}{6}$ cm

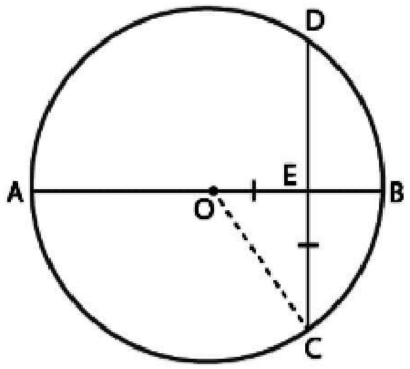
$\Rightarrow r = 7.5$ cm

Hence, the required radius of the circle is 7.5 cm.

Solution 7: AB is the diameter of the circle with centre O, which bisects the chord CD at point E.

Given: CE = ED = 8 cm and EB = 4 cm

Join OC.



Let OC = OB = r cm (Radii of a circle)

Then OE = (r − 4) cm

Now, in right angled ΔOEC, we have:

$OC^2 = OE^2 + EC^2$ (Pythagoras theorem)

$\Rightarrow r^2 = (r-4)^2 + 8^2$

$\Rightarrow r^2 = r^2 - 8r + 16 + 64$

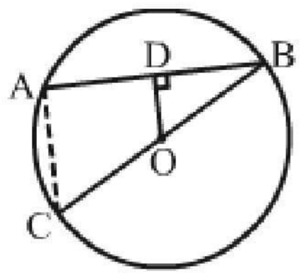
$\Rightarrow r^2 - r^2 + 8r = 80$

$\Rightarrow 8r = 80$

$\Rightarrow r = 10$ cm

Hence, the required radius of the circle is 10 cm.

Solution 8:



Given: BC is a diameter of a circle with centre O and OD ⊥ AB.

To prove: AC parallel to OD and AC = 2 × OD

Construction: Join AC.

Proof:

We know that the perpendicular from the centre of a circle to a chord bisects the chord.

Here, OD ⊥ AB

D is the mid-point of AB.

i.e., AD = BD

Also, O is the mid-point of BC.

i.e., $OC = OB$

Now, in ΔABC, we have:

D is the mid-point of AB and O is the mid-point of BC.

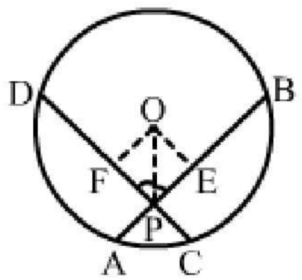
According to the mid-point theorem, the line segment joining the mid points of any two sides of a triangle is parallel to the third side and equal to half of it.

i.e., $OD \parallel AC$ and

$\therefore AC = 2 \times OD$

Hence, proved.

Solution 9:



Given: O is the centre of a circle in which chords AB and CD intersect at P such that PO bisects $\angle BPD$.

To prove: $AB = CD$

Construction: Draw $OE \perp AB$ and $OF \perp CD$

Proof: In ΔOEP and ΔOFP, we have:

$\angle OEP = \angle OFP$ (90° each)

$OP = OP$ (Common)

$\angle OPE = \angle OPF$ ($\because$ OP bisects $\angle BPD$)

Thus, $\Delta OEP \cong \Delta OFP$ (AAS criterion)

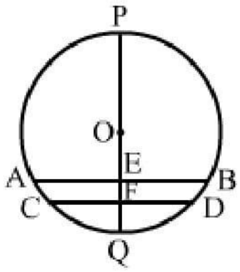
$\Rightarrow OE = OF$

Thus, chords AB and CD are equidistant from the centre O.

$\Rightarrow AB = CD$ ($\because$ Chords equidistant from the centre are equal)

$\therefore AB = CD$

Solution 10:



Given: AB and CD are two parallel chords of a circle with centre O.

POQ is a diameter which is perpendicular to AB.

To prove: $PF \perp CD$ and $CF = FD$

Proof:

AB || CD and POQ is a diameter.

$\angle PEB = 90°$ (Given)

$\angle PFD = \angle PEB$ ($\because$ AB || CD, Corresponding angles)

Thus, $PF \perp CD$

$\therefore OF \perp CD$

We know that the perpendicular from the centre to a chord bisects the chord.

i.e., $CF = FD$

Hence, POQ is perpendicular to CD and bisects it.

Solution 11:

Given: Two distinct circles

To prove: Two distinct circles cannot intersect each other in more than two points.

Proof: Suppose that two distinct circles intersect each other in more than two points.

$\therefore$ These points are non-collinear points.

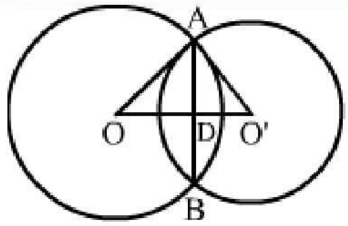
Three non-collinear points determine one and only one circle.

$\therefore$ There should be only one circle.

This contradicts the given, which shows that our assumption is wrong.

Hence, two distinct circles cannot intersect each other in more than two points.

Solution 12:



Given: OA = 10 cm, O'A = 8 cm and AB = 12 cm

$AD = \frac{AB}{2} = \frac{12}{2} = 6$ cm

Now, in right angled ΔADO, we have:

$OA^2 = AD^2 + OD^2$

$\Rightarrow OD^2 = OA^2 - AD^2$

$\Rightarrow OD^2 = 10^2 - 6^2$

$\Rightarrow OD^2 = 100 - 36$

$\Rightarrow OD^2 = 64$

$\therefore OD = 8$ cm

Similarly, in right angled ΔADO', we have:

$O'A^2 = AD^2 + O'D^2$

$\Rightarrow O'D^2 = O'A^2 - AD^2$

$\Rightarrow O'D^2 = 8^2 - 6^2$

$\Rightarrow O'D^2 = 64 - 36$

$\Rightarrow O'D^2 = 28$

$\Rightarrow O'D = \sqrt{28}$

$\Rightarrow O'D = \sqrt{4 \times 7}$

$\Rightarrow O'D = 2\sqrt{7}$ cm

Thus, OO' = (OD + O'D)

$= \left(8 + 2\sqrt{7}\right)$ cm

Hence, the distance between their centres is $\left(8 + 2\sqrt{7}\right)$ cm

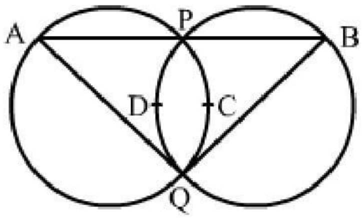
Solution 13:

Given: Two equal circles intersect at point P and Q.

A straight line passes through P and meets the circle at points A and B.

To prove: QA = QB

Construction: Join PQ.



Proof:

Two circles will be congruent if and only if they have equal radii.

Here, PQ is the common chord to both the circles.

Thus, their corresponding arcs are equal (if two chords of a circle are equal, then their corresponding arcs are congruent).

So, arc PCQ = arc PDQ

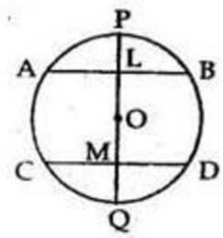
$\therefore \angle QAP = \angle QBP$ (Congruent arcs have the same degree in measure)

Hence, QA = QB (In isosceles triangle, base angles are equal)

Solution 14:

Given: AB and CD are two chords of a circle with centre O. Diameter POQ bisects them at points L and M.

To prove: AB || CD



Proof: AB and CD are two chords of a circle with centre O. Diameter POQ bisects them at L and M.

Then $OL \perp AB$

Also, $OM \perp CD$

$\therefore \angle ALM = \angle LMD = 90^{\circ}$

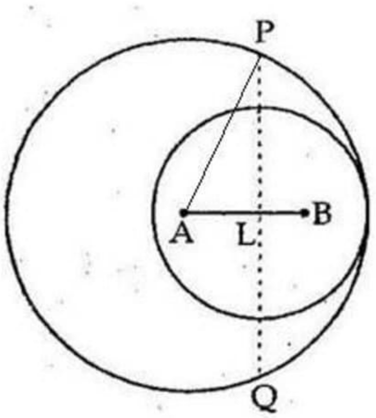
Since alternate angles are equal, we have:

AB|| CD

Solution 15: Two circles with centres A and B of respective radii 5 cm and 3 cm touch each other internally.

The perpendicular bisector of AB meets the bigger circle at P and Q.

Join AP.



Let PQ intersect AB at point L.

Here, AP = 5 cm

Then AB = (5 − 3) cm = 2 cm

Since PQ is the perpendicular bisector of AB, we have:

$AL = \frac{AB}{2} = \frac{2}{2} = 1$ cm

Now, in right angled ΔPLA, we have:

$AP^2 = AL^2 + PL^2$

$\Rightarrow PL^2 = AP^2 - AL^2$

$\Rightarrow PL^2 = 5^2 - 1^2$

$\Rightarrow PL^2 = 25 - 1$

$\Rightarrow PL^2 = 24$

$\Rightarrow PL = \sqrt{24}$

$\Rightarrow PL = \sqrt{4 \times 6}$

$\Rightarrow PL = 2\sqrt{6}$

Thus PQ = 2 × PL

$= (2 \times 2\sqrt{6})$

$= 4\sqrt{6}$ cm

Hence, the required length of PQ is $4\sqrt{6}\ cm$

Solution 16:

We have:

$OB = BC, \angle BOC = \angle BCO = y$

External $\angle OBA = \angle BOC + \angle BCO = (2y)$

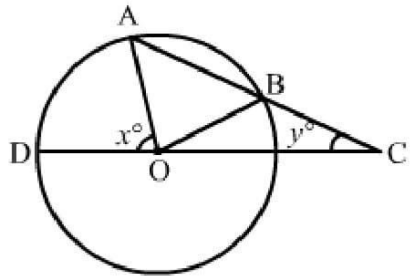
Again, $OA = OB, \angle OAB = \angle OBA = (2y)$

External $\angle AOD = \angle OAC + \angle ACO$

Or $x = \angle OAB + \angle BCO$

Or $x = (2y) + y = 3y$

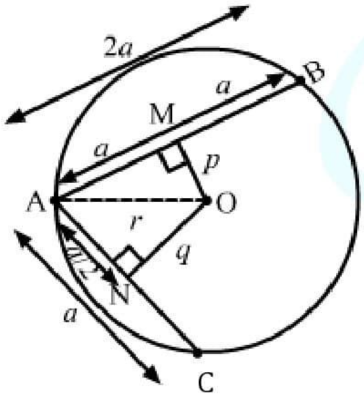
Hence, $x = 3y$



Solution 17:

Let $AC = a$.

Since, $AB = 2AC, \therefore AB = 2a$.



From centre O, perpendicular is drawn to the chords AB and AC at points M and N, respectively.

It is given that $OM = p$ and $ON = q$.

We know that perpendicular drawn from the centre to the chord, bisects the chord.

$\therefore AM = MB = a$ ... (i)

and $AN = NC = \dfrac{a}{2}$ ... (ii)

In ΔOAN,

$(AN)^2 + (NO)^2 = (OA)^2$ (Pythagoras theorem)

$\Rightarrow \left(\frac{a}{2}\right)^2 + q^2 = r^2$

$\Rightarrow \frac{a^2}{4} + q^2 = r^2$

$\Rightarrow \frac{a^2 + 4q^2}{4} = r^2$

$\Rightarrow a^2 + 4q^2 = 4r^2$

$\Rightarrow a^2 = 4r^2 - 4q^2$ ... (iii)

In ΔOAM,

$(AM)^2 + (MO)^2 = (OA)^2$ (Pythagoras theorem)

$\Rightarrow a^2 + p^2 = r^2$

$\Rightarrow a^2 = r^2 - p^2$ ... (iv)

From eq. (iii) and (iv),

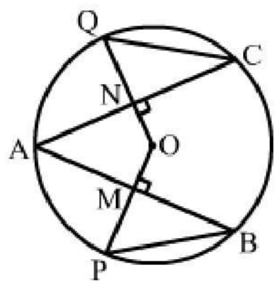
$4r^2 - 4q^2 = r^2 - p^2$

$\Rightarrow 4r^2 - r^2 + p^2 = 4q^2$

$\Rightarrow 3r^2 + p^2 = 4q^2$

Hence, $4q^2 = p^2 + 3r^2$.

Solution 18:



Given: AB and AC are chords of the circle with centre O. AB = AC, OP ⊥ AB and OQ ⊥ AC

To prove: PB = QC

Proof:

AB = AC (Given)

$\Rightarrow \frac{1}{2}AB = \frac{1}{2}AC$

The perpendicular from the centre of a circle to a chord bisects the chord.

$\therefore$ MB = NC ...(i)

Also, OM = ON (Equal chords of a circle are equidistant from the centre)

and OP = OQ (Radii)

$\Rightarrow$ OP - OM = OQ - ON

$\therefore$ PM = QN ...(ii)

Now, in ΔMPB and ΔNQC, we have:

MB = NC [From (i)]

$\angle$PMB = $\angle$QNC [90° each]

PM = QN [From (ii)]

i.e., $\Delta MPB \cong \Delta NQC$ (SAS criterion)

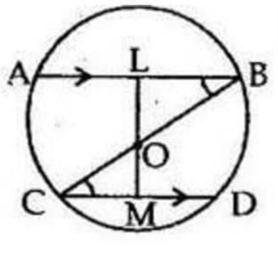
$\therefore$ PB = QC (CPCT)

Solution 19:

Given: BC is a diameter of a circle with centre O. AB and CD are two chords such that AB || CD.

TO prove: AB = CD

Construction: Draw OL $\perp$ AB and OM $\perp$ CD.



Proof:

In ΔOLB and ΔOMC, we have:

$\angle$OLB = $\angle$OMC [90° each]

$\angle$OBL = $\angle$OCD [Alternate angles as AB || CD]

OB = OC [Radii of a circle]

$\therefore \Delta OLB \cong \Delta OMC$ (AAS criterion)

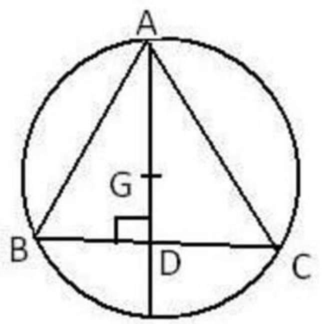
Thus, $OL = OM$ (CPCT)

We know that chords equidistant from the centre are equal.

Hence, $AB = CD$

Solution 20: Let ΔABC be an equilateral triangle of side 9 cm.

Let AD be one of its median.



Then, $AD \perp BC$ (ΔABC is an equilateral triangle)

Also, $BD = \frac{BC}{2} = \frac{9}{2} = 4.5\text{cm}$

In right angled ΔADB, we have:

$AB^2 = AD^2 + BD^2$

$\Rightarrow AD^2 = AB^2 - BD^2$

$\Rightarrow AD = \sqrt{AB^2 - BD^2}$

$$\Rightarrow AD = \sqrt{9^2 - \left(\frac{9}{2}\right)^2}$$

$$\Rightarrow AD = \sqrt{81 - \frac{81}{4}}$$

$$\Rightarrow AD = \sqrt{\frac{324 - 81}{4}}$$

$$\Rightarrow AD = \sqrt{\frac{243}{4}}$$

$\Rightarrow AD = \sqrt{\frac{81 \times 3}{4}}$

$\Rightarrow AD = \frac{9\sqrt{3}}{2}$ cm

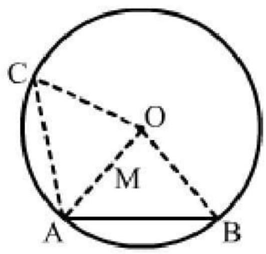
In the equilateral triangle, the centroid and circumcentre coincide and AG : GD = 2 : 1.

Now, radius $= AG = \frac{2}{3}AD$

$\Rightarrow AG = \left(\frac{2}{3} \times \frac{9\sqrt{3}}{2}\right) = 3\sqrt{3}$cm

$\therefore$ The radius of the circle is $3\sqrt{3}$cm

Solution 21: Given: AB and AC are two equal chords of a circle with centre O.



To prove: $\angle OAB = \angle OAC$

Construction: Join OA, OB and OC.

Proof:

In ΔOAB and ΔOAC, we have:

AB = AC (Given)

OA = OA (Common)

OB = OC (Radii of a circle)

$\therefore \Delta\, OAB \cong \Delta\, OAC$ (By SSS congruency rule)

$\Rightarrow \angle OAB = \angle OAC$ (CPCT)

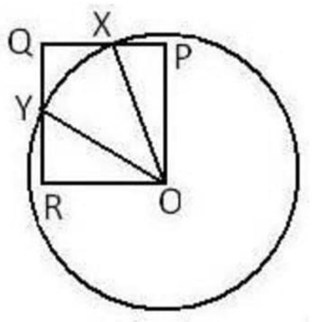
Hence, point O lies on the bisector of $\angle BAC$.

Solution 22:

Given: OPQR is a square. A circle with centre O cuts the square at X and Y.

To prove: QX = QY

Construction: Join OX and OY.



Proof:

In ΔOXP and ΔOYR, we have:

$\angle OPX = \angle ORY$ (90° each)

$OX = OY$ (Radii of a circle)

$OP = OR$ (Sides of a square)

$\therefore \Delta OXP \cong \Delta OYR$ (BY RHS congruency rule)

$\Rightarrow PX = RY$ (By CPCT)

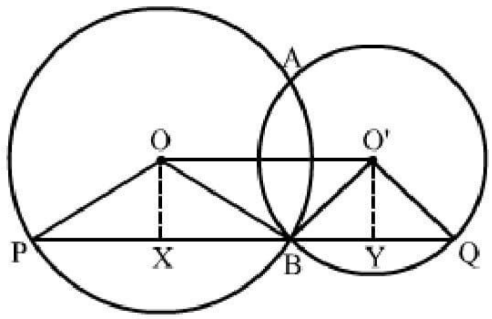
$\Rightarrow PQ - PX = QR - RY$ (PQ and QR are sides of a square)

$\Rightarrow QX = QY$

Hence, proved.

Solution 23:

Given: Two circles with centres O and O' intersect at two points A and B.



Draw a line PQ parallel to OO' through B, OX perpendicular to PQ, O'Y perpendicular to PQ, join all.

We know that perpendicular drawn from the centre to the chord, bisects the chord.

$\therefore PX = XB$ and $YQ = BY$

$\therefore PX + YQ = XB + BY$

On adding $XB + BY$ on both sides, we get

$PX + YQ + XB + BY = 2(XB + BY)$

$\Rightarrow PQ = 2(XY)$

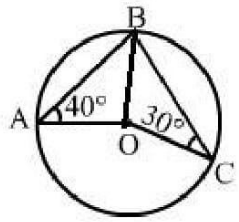
$\Rightarrow PQ = 2(OO')$

Hence, $PQ = 2(OO)'$

Exercise 12B

Solution 1:

(i) Join BO.



In ΔBOC, we have:

$OC = OB$ (Radii of a circle)

$\Rightarrow \angle OBC = \angle OCB$

$\angle OBC = 30°$ ...(i)

In ΔBOA, we have:

$OB = OA$ (Radii of a circle)

$\Rightarrow \angle OBA = \angle OAB$ $[\because \angle OAB = 40°]$

$\Rightarrow \angle OBA = 40°$ ...(ii)

Now, we have:

$\angle ABC = \angle OBC + \angle OBA$

$\Rightarrow \angle ABC = 30° + 40°$ [From (i) and (ii)]

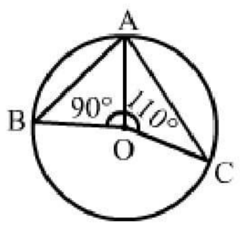
$\therefore \angle ABC = 70°$

The angle subtended by an arc of a circle at the centre is double the angle subtended by the arc at any point on the circumference.

i.e., $\angle AOC = 2\angle ABC$

$= (2 \times 70°) = 140°$

(ii)



Here, $\angle BOC = \{360° - (90° + 110°)\}$

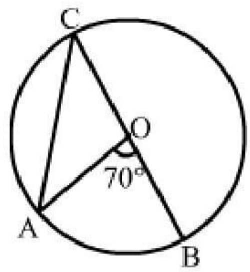
$= (360° - 200°) = 160°$

We know that $\angle BOC = 2\angle BAC$

$\Rightarrow \angle BAC = \dfrac{\angle BOC}{2} = \dfrac{160°}{2} = 80°$

Hence, $\angle BAC = 80°$

Solution 2:



(i) The angle subtended by an arc of a circle at the centre is double the angle subtended by the arc at any point on the circumference.

Thus, $\angle AOB = 2\angle OCA$

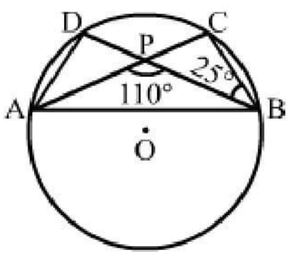
$\Rightarrow \angle OCA = \dfrac{\angle AOB}{2} = \dfrac{70°}{2} = 35°$

(ii) $OA = OC$ (Radii of a circle)

$\angle OAC = \angle OCA$ [Base angles of an isosceles triangle are equal]

$= 35°$

Solution 3: From the given diagram, we have:



$\angle ACB = \angle PCB$

$\angle BPC = (180° - 110°) = 70°$ (Linear pair)

Considering ΔPCB, we have:

$\angle PCB + \angle BPC + \angle PBC = 180^\circ$ (Angle sum property)

$\Rightarrow \angle PCB + 70^\circ + 25^\circ = 180^\circ$

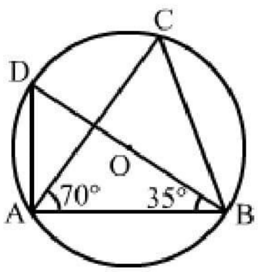
$\Rightarrow \angle PCB = (180^\circ - 95^\circ) = 85^\circ$

$\Rightarrow \angle ACB = \angle PCB = 85^\circ$

We know that the angles in the same segment of a circle are equal.

$\therefore \angle ADB = \angle ACB = 85^\circ$

Solution 4:



It is clear that BD is the diameter of the circle.

Also, we know that the angle in a semicircle is a right angle.

i.e., $\angle BAD = 90^\circ$

Now, considering the ΔBAD, we have:

$\angle ADB + \angle BAD + \angle ABD = 180^\circ$ (Angle sum property of a triangle)

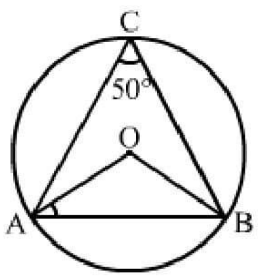
$\Rightarrow \angle ADB + 90^\circ + 35^\circ = 180^\circ$

$\Rightarrow \angle ADB = (180^\circ - 125^\circ) = 55^\circ$

Angles in the same segment of a circle are equal.

Hence, $\angle ACB = \angle ADB = 55^\circ$

Solution 5:



We know that the angle subtended by an arc of a circle at the centre is double the angle subtended by the arc at any point on the circumference.

$\angle AOB = 2\angle ACB$

$\Rightarrow \angle AOB = 2 \times 50°$

$\Rightarrow \angle AOB = 100°$ ...(i)

Let us consider the triangle ΔOAB.

$OA = OB$ (Radii of a circle)

Thus, $\angle OAB = \angle OBA$

In ΔOAB, we have:

$\angle AOB + \angle OAB + \angle OBA = 180°$

$\Rightarrow 100° + \angle OAB + \angle OAB = 180°$ $[\because \angle OAB = \angle OBA]$

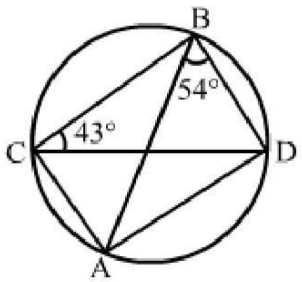
$\Rightarrow 100° + 2\angle OAB = 180°$

$\Rightarrow 2\angle OAB = 180° - 100° = 80°$

$\Rightarrow \angle OAB = 40°$

Hence, $\angle OAB = 40°$

Solution 6:



(i) We know that the angles in the same segment of a circle are equal.

i.e., $\angle ABD = \angle ACD = 54°$

(ii) We know that the angles in the same segment of a circle are equal.

i.e., $\angle BAD = \angle BCD = 43°$

(iii) In ΔABD, we have:

$\angle BAD + \angle ADB + \angle DBA = 180°$ (Angle sum property of a triangle)

$\Rightarrow 43° + \angle ADB + 54° = 180°$

$\Rightarrow \angle ADB = (180° - 97°)$

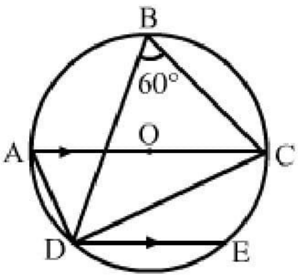
$\Rightarrow \angle ADB = 83°$

$\Rightarrow \angle BDA = 83°$

Solution 7:

Angles in the same segment of a circle are equal.

i.e., $\angle CAD = \angle CBD = 60°$



We know that an angle in a semicircle is a right angle.

i.e., $\angle ADC = 90°$

In ΔADC, we have:

$\angle ACD + \angle ADC + \angle CAD = 180°$ (Angle sum property of a triangle)

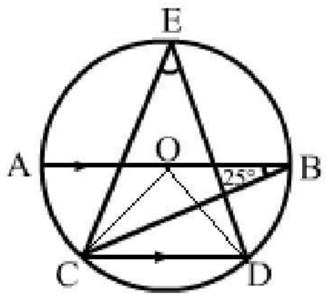
$\Rightarrow \angle ACD + 90° + 60° = 180°$

$\Rightarrow \angle ACD = 180° - (90° + 60°) = (180° - 150°) = 30°$

$\Rightarrow \angle CDE = \angle ACD = 30°$ (Alternate angles as AC parallel to DE)

Hence, $\angle CDE = 30°$

Solution 8:



$\angle BCD = \angle ABC = 25°$ (Alternate angles)

Join CO and DO.

We know that the angle subtended by an arc of a circle at the centre is double the angle subtended by an arc at any point on the circumference.

Thus, $\angle BOD = 2\angle BCD$

$\Rightarrow \angle BOD = 2 \times 25° = 50°$

Similarly, $\angle AOC = 2\angle ABC$

$\Rightarrow \angle AOC = 2 \times 25° = 50°$

AB is a straight line passing through the centre.

i.e., $\angle AOC + \angle COD + \angle BOD = 180°$

$\Rightarrow 50° + \angle COD + 50° = 180°$

$\Rightarrow \angle COD = (180° - 100°) = 80°$

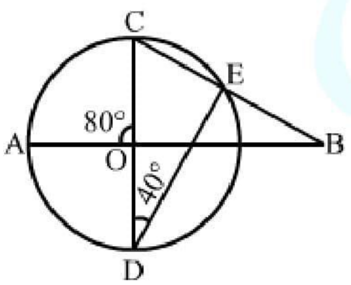
$\Rightarrow \angle CED = \frac{1}{2}\angle COD$

$\Rightarrow \angle CED = \left(\frac{1}{2} \times 80°\right)$

$\Rightarrow \angle CED = 40°$

$\therefore \angle CED = 40°$

Solution 9:



(i) $\angle CED = 90°$ (Angle in a semi-circle)

In ΔCED, we have:

$\angle CED + \angle EDC + \angle DCE = 180°$ (Angle sum property of a triangle)

$\Rightarrow 90° + 40° + \angle DCE = 180°$

$\Rightarrow \angle DCE = (180° - 130°) = 50°$ ...(i)

$\therefore \angle DCE = 50°$

(ii) As $\angle AOC$ and $\angle BOC$ are linear pair, we have:

$\angle BOC = (180° - 80°) = 100°$ ...(ii)

In Δ BOC, we have:

$\angle OBC + \angle OCB + \angle BOC = 180°$ (Angle sum property of a triangle)

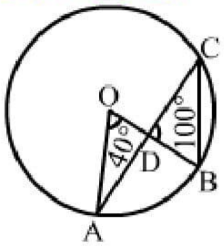
$\Rightarrow \angle ABC + \angle DCE + \angle BOC = 180°$ [$\because \angle OBC = \angle ABC$ and $\angle OCB = \angle DCE$]

$\Rightarrow \angle ABC = 180° - (\angle BOC + \angle DCE)$

$\Rightarrow \angle ABC = 180° - (100° + 50°)$ [From (i) and (ii)]

$\Rightarrow \angle ABC = (180° - 150°) = 30°$

Solution 10:



We know that the angle subtended by an arc of a circle at the centre is double the angle subtended by the arc at any point on the circumference.

$\angle AOB = 2\angle ACB$

$= 2\angle DCB$ [$\because \angle ACB = \angle DCB$]

$\therefore \angle DCB = \frac{1}{2}\angle AOB$

$\Rightarrow \angle DCB = \left(\frac{1}{2} \times 40°\right)$

$\Rightarrow \angle DCB = 20°$

Considering ΔDBC, we have:

$\angle BDC + \angle DCB + \angle DBC = 180°$

$\Rightarrow 100° + 20° + \angle DBC = 180°$

$\Rightarrow \angle DBC = (180° - 120°)$

$\Rightarrow \angle DBC = 60°$

$\Rightarrow \angle OBC = \angle DBC = 60°$

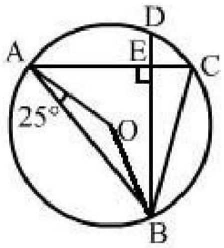
Hence, $\angle OBC = 60°$

Solution 11:

OA = OB (Radii of a circle)

Thus, $\angle OBA = \angle OAB = 25°$

Join OB.



Now in ΔOAB, we have:

$\angle OAB + \angle OBA + \angle AOB = 180°$ (Angle sum property of a triangle)

$\Rightarrow 25° + 25° + \angle AOB = 180°$

$\Rightarrow 50° + \angle AOB = 180°$

$\Rightarrow \angle AOB = (180° - 50°) = 130°$

We know that the angle subtended by an arc of a circle at the centre is double the angle subtended by the arc at any point on the circumference.

i.e., $\angle AOB = 2\angle ACB$

$\Rightarrow \angle ACB = \frac{1}{2}\angle AOB$

$\Rightarrow \angle ACB = \left(\frac{1}{2} \times 130°\right)$

$\Rightarrow \angle ACB = 65°$

Here,

$\angle ACB = \angle ECB$

$\therefore \angle ECB = 65°$ ...(i)

Considering the right angled ΔBEC, we have:

$\angle EBC + \angle BEC + \angle ECB = 180°$ (Angle sum property of a triangle)

$\Rightarrow \angle EBC + 90° + 65° = 180°$ [From(i)]

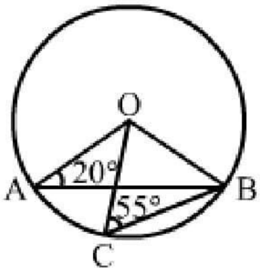
$\Rightarrow \angle EBC = (180° - 155°)$

$\Rightarrow \angle EBC = 25°$

Hence, $\angle EBC = 25°$

Solution 12:

(i)



$OB = OC$ (Radii of a circle)

$\Rightarrow \angle OBC = \angle OCB = 55°$

Considering ΔBOC, we have:

$\angle BOC + \angle OCB + \angle OBC = 180°$ (Angle sum property of a triangle)

$\Rightarrow \angle BOC + 55° + 55° = 180°$

$\Rightarrow \angle BOC = (180° - 110°) = 70°$

(ii) $OA = OB$ (Radii of a circle)

$\Rightarrow \angle OBA = \angle OAB = 20°$

Considering ΔAOB, we have:

$\angle AOB + \angle OAB + \angle OBA = 180°$ (Angle sum property of a triangle)

$\Rightarrow \angle AOB + 20° + 20° = 180°$

$\Rightarrow \angle AOB = (180° - 40°) = 140°$

$\therefore \angle AOC = \angle AOB - \angle BOC$

$= (140° - 70°)$

$= 70°$

Hence, $\angle AOC = 70°$

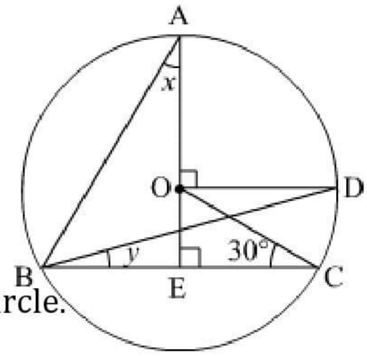
Solution 13: In the given figure, OD is parallel to BC.

$\therefore \angle BCO = \angle COD$ (Alternate interior angles)

$\Rightarrow \angle COD = 30°$ ...(i)

We know that the angle subtended by an arc of a circle at the centre is double the angle subtended by it on the remaining part of the circle.

Here, arc CD subtends $\angle COD$ at the centre and $\angle CBD$ at B on the circle.



$\therefore \angle COD = 2\angle CBD$

$\Rightarrow \angle CBD = \dfrac{30}{2} = 15°$ (from (i))

$\therefore y = 15°$ ...(ii)

Also, arc AD subtends $\angle AOD$ at the centre and $\angle ABD$ at B on the circle.

$\therefore \angle AOD = 2\angle ABD$

$$\Rightarrow \angle ABD = \frac{90°}{2}$$

$\Rightarrow \angle ABD = 45°$ ...(iii)

In ΔABE,

$x + y + \angle ABD + \angle AEB = 180°$ (Sum of the angles of a triangle)

$\Rightarrow x + 15° + 45° + 90° = 180°$ (from (ii) and (iii))

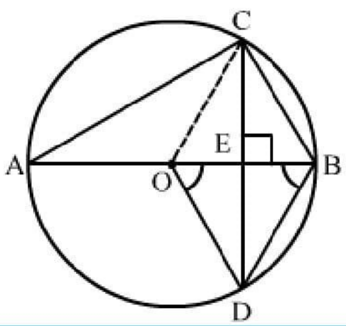
$\Rightarrow x = 180° - (90° + 15° + 45°)$

$\Rightarrow x = 180° - 150°$

$\Rightarrow x = 30°$

Hence, $x = 30°$ and $y = 15°$.

Solution 14: In the given figure, $BD = OD$ and $CD \perp AB$.



Join AC and OC.

In ΔODE and ΔDBE,

$\angle DOE = \angle DBE$ (given)

$\angle DEO = \angle DEB = 90^\circ$

$OD = DB$ (given)

$\therefore$ By AAS congruence rule,

$\Delta ODE \cong \Delta BDE$,

Thus, $OE = EB$ ...(i)

Now, in ΔCOE and ΔCBE,

$CE = CE$ (common)

$\angle CEO = \angle CEB = 90^\circ$

$OE = EB$ (from (1))

$\therefore$ By SAS congruence rule,

$\Delta COE \cong \Delta CBE$,

Thus, $CO = CB$ ...(ii)

Also, $CO = OB = OA$ (radius of the circle) ...(iii)

From (ii) and (iii),

$CO = CB = OB$

$\therefore$ ΔCOB is equilateral triangle.

$\therefore \angle COB = 60^\circ$ ...(iv)

We know that the angle subtended by an arc of a circle at the centre is double the angle subtended by it on the remaining part of the circle.

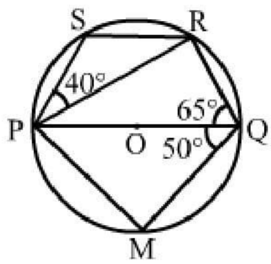
Here, arc CB subtends $\angle COB$ at the centre and $\angle CAB$ at A on the circle.

$\therefore \angle COB = 2\angle CAB$

$\Rightarrow \angle CAB = \dfrac{60°}{2} = 30°$ (from (iv))

Hence, $\angle CAB = 30°$.

Solution 15:



Here, PQ is the diameter and the angle in a semicircle is a right angle.

i.e., $\angle PRQ = 90°$

In ΔPRQ, we have:

$\angle QPR + \angle PRQ + \angle PQR = 180°$ (Angle sum property of a triangle)

$\Rightarrow \angle QPR + 90° + 65° = 180°$

$\Rightarrow \angle QPR = (180° - 155°) = 25°$

In ΔPQM, PQ is the diameter.

$\therefore \angle PMQ = 90°$

In ΔPQM, we have:

$\angle QPM + \angle PMQ + \angle PQM = 180°$ (Angle sum property of a triangle)

$\Rightarrow \angle QPM + 90° + 50° = 180°$

$\Rightarrow \angle QPM = (180° - 140°) = 40°$

Now, in quadrilateral PQRS, we have:

$\angle QPS + \angle SRQ = 180°$ (Opposite angles of a cyclic quadrilateral)

$\Rightarrow \angle QPR + \angle RPS + \angle PRQ + \angle PRS = 180°$

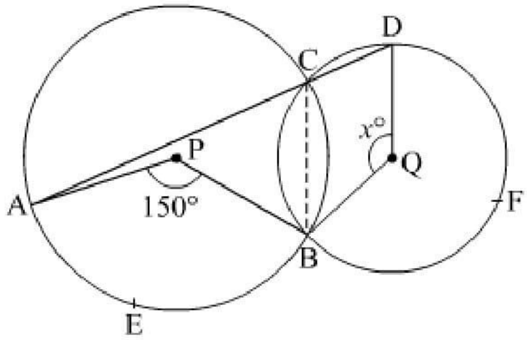
$\Rightarrow 25° + 40° + 90° + \angle PRS = 180°$

$\Rightarrow \angle PRS = 180° - 155° = 25°$

$\therefore \angle PRS = 25°$

Thus, $\angle QPR = 25°$; $\angle QPM = 40°$; $\angle PRS = 25°$

Solution 16: We know that the angle subtended by an arc of a circle at the centre is double the angle subtended by it on the remaining part of the circle.



Here, arc AEB subtends $\angle APB$ at the centre and $\angle ACB$ at C on the circle.

$\therefore \angle APB = 2\angle ACB$

$\Rightarrow \angle ACB = \dfrac{150°}{2} = 75° \qquad \ldots(i)$

Since ACD is a straight line, $\angle ACB + \angle BCD = 180°$

$\Rightarrow \angle BCD = 180° - 75°$

$\Rightarrow \angle BCD = 105° \qquad$ ...(ii)

Also, arc BFD subtends reflex $\angle BQD$ at the centre and $\angle BCD$ at C on the circle.

$\therefore$ reflex $\angle BQD = 2\angle BCD$

$\Rightarrow$ reflex $\angle BQD = 2(105°)$

$\Rightarrow$ reflex $\angle BQD = 210° \qquad$ ...(iii)

Now,

reflex $\angle BQD + \angle BQD = 360°$

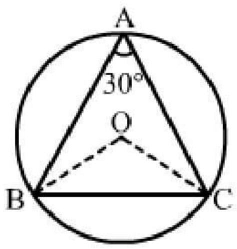
$\Rightarrow 210° + x = 360°$

$\Rightarrow x = 360° - 210°$

$\Rightarrow x = 150°$

Hence, $x = 150^{\circ}$.

Solution 17:



Join OB and OC.

$\angle BOC = 2\angle BAC$ (As angle subtended by an arc of a circle at the centre is double the angle subtended by the arc at any point on the circumference)

$= 2 \times 30^{\circ}$ $\quad [\because \angle BAC = 30^{\circ}]$

$= 60^{\circ}$ $\quad$ ...(i)

Consider ΔBOC, we have:

$OB = OC$ $\quad$ [Radii of a circle]

$\Rightarrow \angle OBC = \angle OCB$ $\quad$ ...(ii)

In ΔBOC, we have:

$\angle BOC + \angle OBC + \angle OCB = 180$ $\quad$ (Angle sum property of a triangle)

$\Rightarrow 60^{\circ} + \angle OCB + \angle OCB = 180^{\circ}$ $\quad$ [From (i) and (ii)]

$\Rightarrow 2\angle OCB = (180^{\circ} - 60^{\circ}) = 120^{\circ}$

$\Rightarrow \angle OCB = 60^{\circ}$ $\quad$ ...(ii)

Thus, we have:

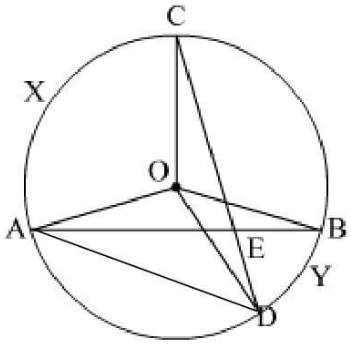
$\angle OBC = \angle OCB = \angle BOC = 60^{\circ}$

Hence, ΔBOC is an equilateral triangle.

i.e., $OB = OC = BC$

$\therefore$ BC is the radius of the circumcircle.

Solution 18:



Join AD

We know that the angle subtended by an arc of a circle at the centre is double the angle subtended by it on the remaining part of the circle.

Here, arc AXC subtends $\angle AOC$ at the centre and $\angle ADC$ at D on the circle.

$\therefore \angle AOC = 2\angle ADC$

$\Rightarrow \angle ADC = \frac{1}{2}\angle AOC \quad \ldots(i)$

Also, arc DYB subtends $\angle DOB$ at the centre and $\angle DAB$ at A on the circle.

$\therefore \ \angle DOB \ = \ 2\angle DAB$

$\Rightarrow \angle DAB = \frac{1}{2}\angle DOB \quad \ldots(ii)$

Now, in ΔADE,

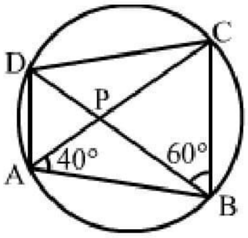
$\angle AEC = \angle ADC + \angle DAB \quad$ (Exterior angle)

$\Rightarrow \angle AEC = \frac{1}{2}(\angle AOC + \angle DOB) \quad$ (from (i) and (ii))

Hence, $\angle AEC \ = \ 1/2$(angle subtended by arc CXA at the centre + angle subtended by arc DYB at the centre).

Exercise 12C

Solution 1:



(i) $\angle BDC = \angle BAC = 40°$ (Angles in the same segment)

In ΔBCD, we have:

$\angle BCD + \angle DBC + \angle BDC = 180°$ (Angle sum property of a triangle)

$\Rightarrow \angle BCD + 60° + 40° = 180°$

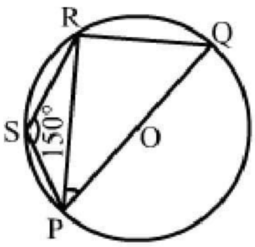
$\Rightarrow \angle BCD = (180° - 100°)$

$\Rightarrow \angle BCD = 80°$

(ii) $\angle CAD = \angle CBD$ (Angles in the same segment)

$= 60°$

Solution 2:



In cyclic quadrilateral PQRS, we have:

$\angle PSR + \angle PQR = 180°$

$\Rightarrow 150° + \angle PQR = 180°$

$\Rightarrow \angle PQR = (180° – 150°)$

$\therefore \angle PQR = 30°$ ...(i)

Also, $\angle PRQ = 90°$ (Angle in a semicircle) ...(ii)

Now, in ΔPRQ, we have:

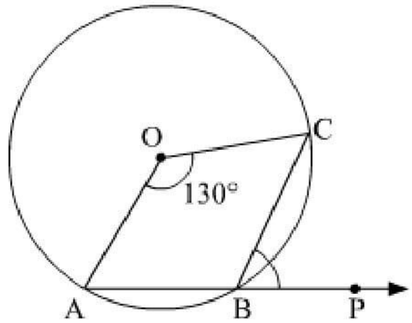
$\angle PQR + \angle PRQ + \angle RPQ = 180°$

$\Rightarrow 30° + 90° + \angle RPQ = 180°$ [From(i) and (ii)]

$\Rightarrow \angle RPQ = 180° - 120°$

$\therefore \angle RPQ = 60°$

Solution 3:



Reflex $\angle AOC + \angle AOC = 360°$

$\Rightarrow$ Reflex $\angle AOC + 130° + x = 360°$

$\Rightarrow$ Reflex $\angle AOC = 360° - 130°$

$\Rightarrow$ Reflex $\angle AOC = 230°$

We know that the angle subtended by an arc of a circle at the centre is double the angle subtended by it on the remaining part of the circle.

Here, arc AC subtends reflex $\angle AOC$ at the centre and $\angle ABC$ at B on the circle.

$\therefore \angle AOC = 2\angle ABC$

$\Rightarrow \angle ABC = \dfrac{230°}{2}$

$\Rightarrow \angle ABC = 115° \qquad ...(i)$

Since ABP is a straight line,

$\angle ABC + \angle PBC = 180°$

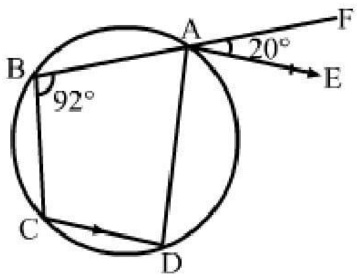
$\angle PBC = 180° - \angle ABC$

$\Rightarrow \angle PBC = 180° - 115° \qquad$ [using(i)]

$\Rightarrow \angle PBC = 65° \qquad ..$

Hence, $\angle PBC = 65°$.

Solution 4:



Given: ABCD is a cyclic quadrilateral.

Then $\angle ABC + \angle ADC = 180°$

$\Rightarrow 92° + \angle ADC = 180°$

$\Rightarrow \angle ADC = (180° - 92°)$

$\Rightarrow \angle ADC = 88°$

Again, AE parallel to CD.

Thus, $\angle EAD = \angle ADC = 88°$ (Alternate angles)

We know that the exterior angle of a cyclic quadrilateral is equal to the interior opposite angle.

$\therefore \angle BCD = \angle DAF$

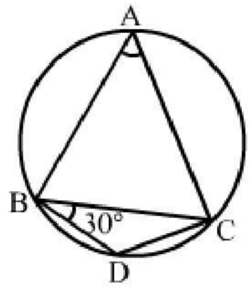
$\Rightarrow \angle BCD = \angle EAD + \angle EAF$

$\Rightarrow \angle BCD = 88° + 20°$

$\Rightarrow \angle BCD = 108°$

Hence, $\angle BCD = 108°$

Solution 5: Given,



$BD = DC$

$\Rightarrow \angle BCD = \angle CBD = 30°$

In ΔBCD, we have:

$\angle BCD + \angle CBD + \angle CDB = 180°$ (Angle sum property of a triangle)

$\Rightarrow 30° + 30° + \angle CDB = 180°$

$\Rightarrow \angle CDB = (180° - 60°) = 120°$

The opposite angles of a cyclic quadrilateral are supplementary.

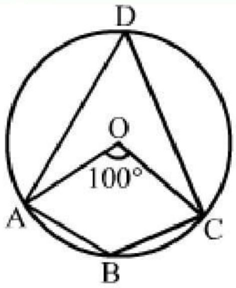
Thus, $\angle CDB + \angle BAC = 180°$

$\Rightarrow 120° + \angle BAC = 180°$

$\Rightarrow \angle BAC = (180° - 120°) = 60°$

$\therefore \angle BAC = 60°$

Solution 16:



We know that the angle subtended by an arc is twice the angle subtended by it on the circumference in the alternate segment.

Thus, $\angle AOC = 2\angle ADC$

$\Rightarrow 100° = 2\angle ADC$

$\therefore \angle ADC = 50°$

The opposite angles of a cyclic quadrilateral are supplementary and ABCD is a cyclic quadrilateral.

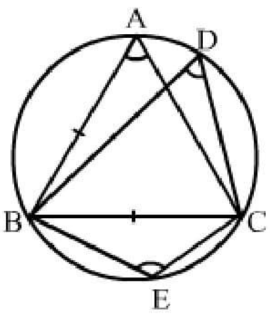
Thus, $\angle ADC + \angle ABC = 180°$

$\Rightarrow 50° + \angle ABC = 180°$

$\Rightarrow \angle ABC = (180° - 50°) = 130°$

$\therefore \angle ADC = 50°$ and $\angle ABC = 130°$

Solution 7:



(i) Given: ΔABC is an equilateral triangle.

i.e., each of its angle = 60°

$\Rightarrow \angle BAC = \angle ABC = \angle ACB = 60°$

Angles in the same segment of a circle are equal.

i.e., $\angle BDC = \angle BAC = 60°$

$\therefore \angle BDC = 60°$

(ii) The opposite angles of a cyclic quadrilateral are supplementary.

Then in cyclic quadrilateral ABEC, we have:

$\angle BAC + \angle BEC = 180°$

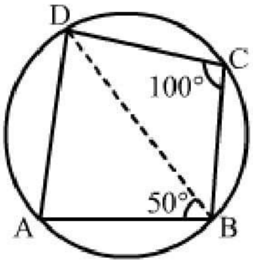
$\Rightarrow 60° + \angle BEC = 180°$

$\Rightarrow \angle BEC = (180° - 60°)$

$\Rightarrow \angle BEC = 120°$

$\therefore \angle BDC = 60°$ and $\angle BEC = 120°$

Solution 8:



Given: ABCD is a cyclic quadrilateral.

$\therefore \angle DAB + \angle DCB = 180°$ (Opposite angles of a cyclic quadrilateral are supplementary)

$\Rightarrow \angle DAB + 100° = 180°$

$\Rightarrow \angle DAB = (180° - 100°) = 80°$

Now, in ΔABD, we have:

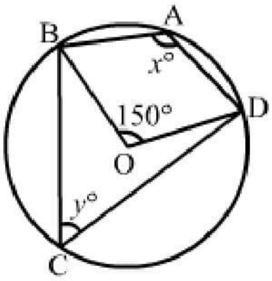
$\Rightarrow \angle DAB + \angle ABD + \angle ADB = 180°$

$\Rightarrow 80° + 50° + \angle ADB = 180°$

$\Rightarrow \angle ADB = (180° - 130°) = 50°$

Hence, $\angle ADB = 50°$

Solution 9:



O is the centre of the circle and $\angle BOD = 150°$.

Thus, reflex angle $\angle BOD = (360° - 150°) = 210°$

Now,

$$x = \frac{1}{2}(\text{reflex}\angle BOD)$$

$$\Rightarrow x = \left(\frac{1}{2} \times 210°\right)$$

$$\Rightarrow x = 105°$$

Again, $x + y = 180°$ (Opposite angles of a cyclic quadrilateral)

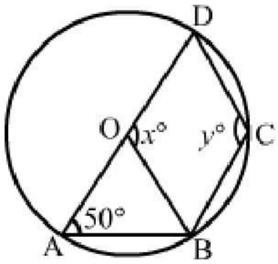
$\Rightarrow 105° + y = 180°$

$\Rightarrow y = (180° - 105°)$

$\therefore y = 75°$

Hence, $x = 105°$ and $y = 75°$

Solution 10:



O is the centre of the circle and $\angle DAB = 50°$.

$OA = OB$ (Radii of a circle)

$\Rightarrow \angle OBA = \angle OAB = 50°$

In ΔOAB, we have:

$\angle OAB + \angle OBA + \angle AOB = 180°$

$\Rightarrow 50° + 50° + \angle AOB = 180°$

$\Rightarrow \angle AOB = (180° - 100°)$

$\Rightarrow \angle AOB = 80°$

Since AOD is a straight line, we have:

$\therefore x = 180° - \angle AOB$

$= (180° - 80°)$

$= 100°$

i.e., $x = 100°$

The opposite angles of a cyclic quadrilateral are supplementary.

ABCD is a cyclic quadrilateral.

Thus,

$\angle DAB + \angle BCD = 180°$

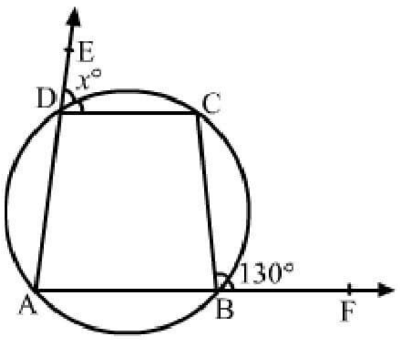
$\Rightarrow \angle BCD = (180° - 50°)$

$\Rightarrow \angle BCD = 130°$

$\therefore y = 130°$

Hence, $x = 100°$ and $y = 130°$

Solution 11:



ABCD is a cyclic quadrilateral.

We know that in a cyclic quadrilateral, the exterior angle is equal to the interior opposite angle.

$\therefore \angle CBF = \angle CDA$

$\Rightarrow \angle CBF = (180° - x)$

$\Rightarrow 130° = 180° - x \quad [\because \angle CBF = 130°]$

$\Rightarrow x = (180° - 130°)$

$\Rightarrow x = 50°$

Hence, $x = 50°$

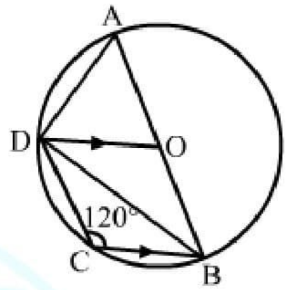
Solution 12: We have,

AB is a diameter of the circle where O is the centre, DO || BC and $\angle BCD = 120°$.

(i) Since ABCD is a cyclic quadrilateral, we have:

$\angle BCD + \angle BAD = 180°$

$\Rightarrow 120° + \angle BAD = 180°$



$\Rightarrow \angle BAD = (180° - 120°)$

$\therefore \angle BAD = 60°$

(ii) $\angle BDA = 90°$ (Angle in a semicircle)

In Δ ABD, we have:

$\angle BDA + \angle BAD + \angle ABD = 180°$

$\Rightarrow 90° + 60° + \angle ABD = 180°$

$\Rightarrow \angle ABD = (180° - 150°)$

$\therefore \angle ABD = 30°$

(iii) OD = OA (Radii of a circle)

$\angle ODA = \angle OAD$

$= \angle BAD$

$= 60°$

$\angle ODB = 90° - \angle ODA$

$= (90° - 60°)$

$= 30°$

Here, DO || BC (Given; alternate angles)

$\angle CBD = \angle ODB = 30°$

$\therefore \angle CBD = 30°$

(iv) $\angle ADC = \angle ADB + \angle CDB$

$= 90° + 30°$

$= 120°$

In ΔAOD, we have:

$\angle ODA + \angle OAD + \angle AOD = 180°$

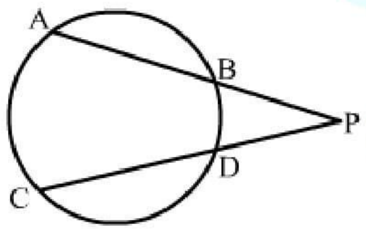
$\Rightarrow 60° + 60° + \angle AOD = 180°$

$\Rightarrow \angle AOD = 180° - 120°$

$\Rightarrow \angle AOD = 60°$

Since all the angles of ΔAOD are of $60°$ each, ΔAOD is an equilateral triangle.

Solution 13:



AB and CD are two chords of a circle which intersect each other at P outside the circle.

AB = 6 cm, BP = 2 cm and PD = 2.5 cm

$\therefore AP \times BP = CP \times DP$

$\Rightarrow 8 \times 2 = (CD + 2.5) \times 2.5$ [$\because CP = CD + DP$]

Let CD = x cm

Thus, $8 \times 2 = (CD + 2.5) \times 2.5$

$\Rightarrow 16 = 2.5x + 6.25$

$\Rightarrow 2.5x = (16 - 6.25)$

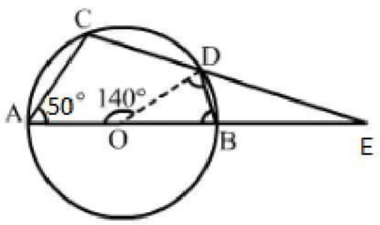
$\Rightarrow 2.5x = 9.75$

$\Rightarrow x = \frac{9.75}{2.5}$

$\Rightarrow x = 3.9$

Hence, CD = 3.9 cm

Solution 14:



O is the centre of the circle where $\angle AOD = 140°$ and $\angle CAB = 50°$.

(i) $\angle BOD = 180° - \angle AOD$

$= (180° - 140°)$

$= 40°$

We have the following:

OB = OD (Radii of a circle)

$\angle OBD = \angle ODB$

In ΔOBD, we have:

$\angle BOD + \angle OBD + \angle ODB = 180°$

$\Rightarrow \angle BOD + \angle OBD + \angle OBD = 180°$ $[\because \angle OBD = \angle ODB]$

$\Rightarrow 40° + 2\angle OBD = 180°$

$\Rightarrow 2\angle OBD = (180° - 40°)$

$\Rightarrow 2\angle OBD = 140°$

$\Rightarrow \angle OBD = 70°$

Since ABCD is a cyclic quadrilateral, we have:

$\angle CAB + \angle BDC = 180°$

$\Rightarrow \angle CAB + \angle ODB + \angle ODC = 180°$

$\Rightarrow 50° + 70° + \angle ODC = 180°$

$\Rightarrow \angle ODC = (180° - 120°)$

$\therefore \angle ODC = 60°$

$\angle EDB = (180° - (\angle ODC + \angle ODB)$

$= 180° - (60° + 70°)$

$= 180° - 130°$

$= 50°$

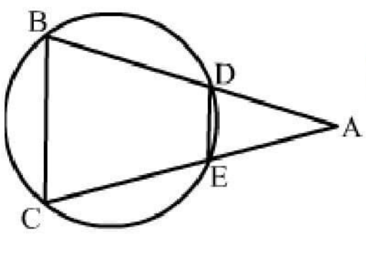
$\therefore \angle EDB = 50°$

(ii) $\angle EBD = 180° - \angle OBD$

$= 180° - 70°$

$= 110°$

Solution 15:



ABC is an isosceles triangle.

Here, AB = AC

$\therefore \angle ACB = \angle ABC$ ...(i)

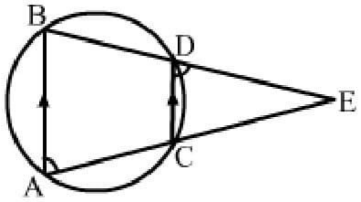
So, exterior $\angle ADE = \angle ACB$

$= \angle ABC$ [from(i)]

$\therefore \angle ADE = \angle ABC$ (Corresponding angles)

Hence, DE || BC

Solution 16:



AB and CD are two parallel chords of a circle. BDE and ACE are two straight lines that intersect at E.

If one side of a cyclic quadrilateral is produced, then the exterior angle is equal to the interior opposite angle.

$\therefore$ Exterior $\angle EDC = \angle A$ ...(i)

Exterior $\angle DCE = \angle B$ ...(ii)

Also, AB parallel to CD.

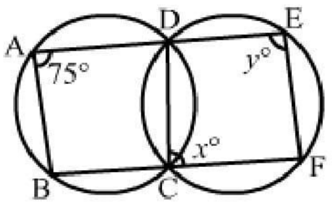
Then, $\angle EDC = \angle B$ (Corresponding angles)

and $\angle DCE = \angle A$ (Corresponding angles)

$\therefore \angle A = \angle B$ [From(i) and (ii)]

Hence, ΔAEB is isosceles.

Solution 17:



We know that if one side of a cyclic quadrilateral is produced, then the exterior angle is equal to the interior opposite angle.

i.e., $\angle BAD = \angle DCF = 75°$

$\Rightarrow \angle DCF = x = 75°$

Again, the sum of opposite angles in a cyclic quadrilateral is 180°.

Thus, $\angle DCF + \angle DEF = 180°$

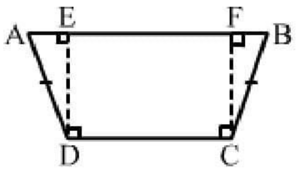
$\Rightarrow 75° + y = 180°$

$\Rightarrow y = (180° - 75°)$

$\Rightarrow y = 105°$

Hence, $x = 75°$ and $y = 105°$

Solution 18:



ABCD is a quadrilateral in which AD = BC and $\angle ADC = \angle BCD$.

Draw $DE \perp AB$ and $CF \perp AB$.

In ΔADE and ΔBCF, we have:

$\angle ADE = \angle ADC - 90° = \angle BCD - 90° = \angle BCF$ (Given: $\angle ADC = \angle BCD$)

AD = BC (Given)

and $\angle AED = \angle BCF = 90°$

$\therefore \Delta ADE \cong \Delta BCF$ (By AAS congruency)

$\Rightarrow \angle A = \angle B$

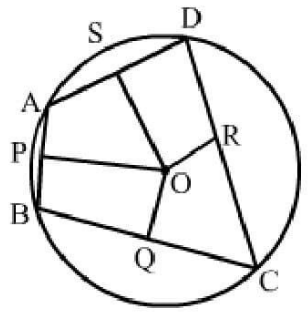
Now, $\angle A + \angle B + \angle C + \angle D = 360°$

$\Rightarrow 2\angle B + 2\angle D = 360°$

$\Rightarrow \angle B + \angle D = 180°$

Hence, ABCD is a cyclic quadrilateral.

Solution 19: Let ABCD be the cyclic quadrilateral and PO, QO, RO and SO be the perpendicular bisectors of sides AB, BC, CD and AD.



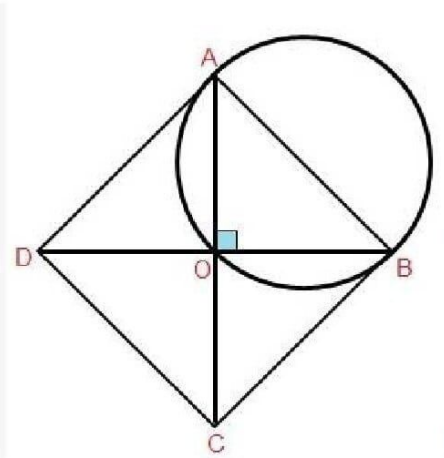
We know that the perpendicular bisector of a chord passes through the centre of the circle.

Since, AB, BC, CD and AD are the chords of a circle, PO, QO, RO and SO pass through the centre.

i.e., PO, QO, RO and SO are concurrent.

Hence, the perpendicular bisectors of the sides of a cyclic quadrilateral are concurrent.

Solution 20: Let ABCD be the rhombus with AC and BD as diagonals intersecting at point O.



The diagonals of a rhombus bisect each other at right angles.

i.e., $\angle AOB = \angle BOC = \angle COD = \angle AOD = 90°$

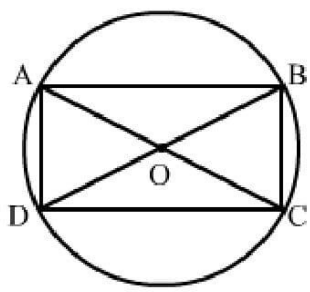
Now, circles with AB, BC, CD and DA as diameter passes through O (angle in a semi-circle is a right angle).

Hence, the circle with four sides of a rhombus as diameter, pass through O, i.e., the point of intersection of its diagonals.

Solution 21:

Given: ABCD is a cyclic rectangle whose diagonals intersect at O.

To prove: O is the centre of the circle.



Proof:

Here, $\angle BCD = 90°$ [Since it is a rectangle]

So, BD is the diameter of the circle (if the angle made by the chord at the circle is right angle, then the chord is the diameter).

Also, diagonals of a rectangle bisect each other and are equal.

$\therefore OA = OB = OC = OD$

BD is the diameter.

$\therefore$ BO and OD are the radius.

Thus, O is the centre of the circle.

Also, the centre of the circle is circumscribing the cyclic rectangle.

Hence, O is the point of intersection of the diagonals of ABCD.

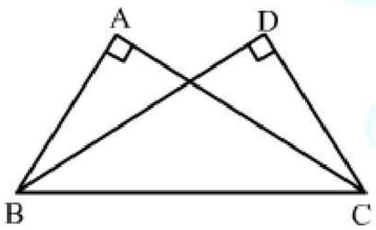
Solution 22: Let A, B and C be the given points.

With B as the centre and a radius equal to AC, draw an arc.

With C as the centre and AB as radius, draw another arc intersecting the previous arc at D.

Then D is the desired point.

Proof: Join BD and CD.



In ΔABC and ΔDCB, we have:

$AB = DC$

$AC = DB$

$BC = CB$

i.e., $\Delta ABC \cong \Delta DCB$

$\Rightarrow \angle BAC = \angle CDB$

Thus, BC subtends equal angles $\angle BAC$ and $\angle CDB$ on the same side of it.

$\therefore$ Points A, B, C and D are cyclic.

Solution 23:

In cyclic quadrilateral ABCD, we have:

$\angle B + \angle D = 180°$ ...(i) (Opposite angles of a cyclic quadrilateral)

$\angle B - \angle D = 60°$ ...(ii) (Given)

From (i) and (ii), we get:

$2\angle B = 240°$

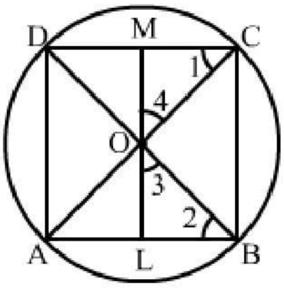
$\Rightarrow \angle B = 120°$

$\therefore \angle D = 60°$

Hence, the smaller of the two angles is 60°.

Solution 24: Let ABCD be a cyclic quadrilateral whose diagonals AC and BD intersect at O at right angles.

Let $OL \perp AB$ such that LO produced meets CD at M.



Then we have to prove that CM = MD

Clearly, $\angle 1 = \angle 2$ [Angles in the same segment]

$\angle 2 + \angle 3 = 90°$ [$\because \angle OLB = 90°$]

$\angle 3 + \angle 4 = 90°$ [$\because$ LOM is a straight line and $\angle BOC = 90°$]

$\therefore \angle 2 + \angle 3 = \angle 3 + \angle 4$

$\Rightarrow \angle 2 = \angle 4$

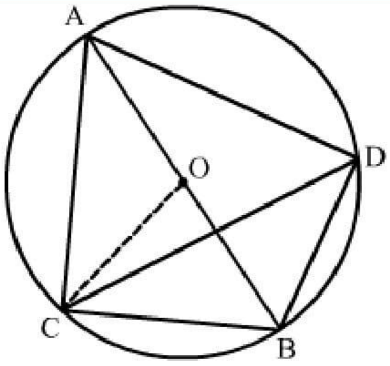
Thus, $\angle 1 = \angle 2$ and $\angle 2 = \angle 4$

$\Rightarrow \angle 1 = \angle 4$

$\therefore$ OM = CM and, similarly, OM = MD

Hence, CM = MD

Solution 25: Draw two right triangles ACB and ADB in a circle with centre O, where AB is the diameter of the circle.



Join CO.

We know that the angle subtended by an arc of a circle at the centre is double the angle subtended by it on the remaining part of the circle.

Here, arc CB subtends $\angle COB$ at the centre and $\angle CAB$ at A on the circle.

$\therefore \angle COB = 2\angle CAB$ ...(i)

Also, arc CB subtends $\angle COB$ at the centre and $\angle CDB$ at D on the circle.

$\therefore \angle COB = 2\angle CDB$ ...(ii)

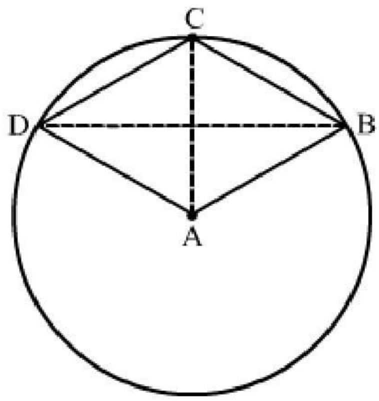
Equating (i) and (ii),

$2\angle CAB = 2\angle CDB$

$\Rightarrow \angle CAB = \angle CDB$

Hence, $\angle BAC = \angle BDC$.

Solution 26: In the given figure, ABCD is a quadrilateral such that A is the centre of the circle passing through B, C and D.



Join AC and BD.

We know that the angle subtended by an arc of a circle at the centre is double the angle subtended by it on the remaining part of the circle.

Here, arc CD subtends $\angle CAD$ at the centre and $\angle CBD$ at B on the circle.

$\therefore \angle CAD = 2\angle CBD$ ...(i)

Also, arc CB subtends $\angle CAB$ at the centre and $\angle CDB$ at D on the circle.

$\therefore \angle CAB = 2\angle CDB$ ...(ii)

Adding (i) and (ii), we get

$\angle CAD + \angle CAB = 2(\angle CBD + \angle CDB)$

$\Rightarrow \angle BAD = 2(\angle CBD + \angle CDB)$

$\Rightarrow \angle CBD + \angle CDB = \frac{1}{2} \angle BAD$

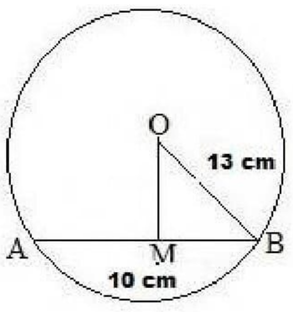
Hence, $\angle CBD + \angle CDB = \frac{1}{2} \angle BAD$.

Multiple Choice Questions (MCQ)

Solution 1: (b) 12 cm

Let AB be the chord of the given circle with centre O and a radius of 13 cm.

Then, AB = 10 cm and OB = 13 cm



From O, draw OM perpendicular to AB.

We know that the perpendicular from the centre of a circle to a chord bisects the chord.

$\therefore$ BM $= \frac{10}{2}$ cm $=$ 5cm

From the right ΔOMB, we have:

$OB^2 = OM^2 + MB^2$

$\Rightarrow 13^2 = OM^2 + 5^2$

$\Rightarrow 169 = OM^2 + 25$

$\Rightarrow OM^2 = (169 - 25)$

$\Rightarrow OM^2 = 144$

$\Rightarrow$ OM = 12cm

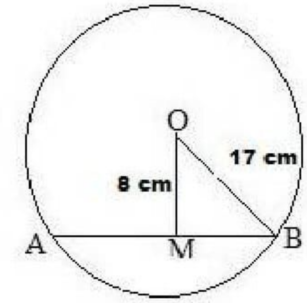
Hence, the distance of the chord from the centre is 12 cm.

Solution 2: (c) 30 cm

Let AB be the chord of the given circle with centre O and a radius of 17 cm.

From O, draw OM perpendicular to AB.

Then OM = 8 cm and OB = 17 cm



From the right ΔOMB, we have:

$OB^2 = OM^2 + MB^2$

$\Rightarrow 17^2 = 8^2 + MB^2$

$\Rightarrow 289 = 64 + MB^2$

$\Rightarrow MB^2 = (289 - 64)$

$\Rightarrow MB^2 = 225$

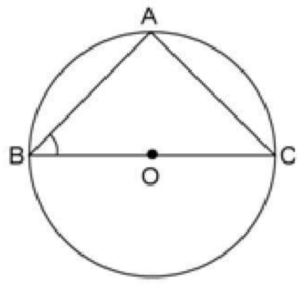
$\Rightarrow MB = 15cm$

The perpendicular from the centre of a circle to a chord bisects the chord.

$\therefore AB = 2 \times MB = (2 \times 15)$ cm = 30 cm

Hence, the required length of the chord is 30 cm.

Solution 3: (b) 45°



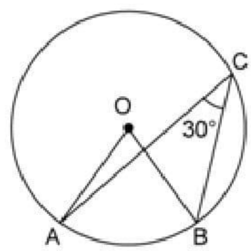
Since an angle in a semicircle is a right angle, $\angle BAC = 90°$

$\therefore \angle ABC + \angle ACB = 90°$

Now, AB = AC (Given)

$\Rightarrow \angle ABC = \angle ACB = 45°$

Solution 4: (c) 60°

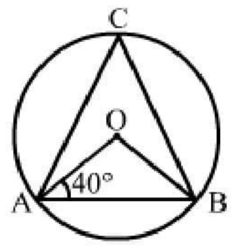


We know that the angle at the centre of a circle is twice

the angle at any point on the remaining part of the circumference.

Thus, $\angle AOB = (2 \times \angle ACB) = (2 \times 30°) = 60°$

Solution 5: (b) 50°



OA = OB

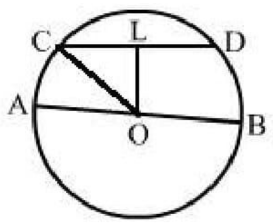
$\Rightarrow \angle OBA = \angle OAB = 40°$

Now, $\angle AOB = 180° - (40° + 40°) = 100°$

$\therefore \angle ACB = \frac{1}{2}\angle AOB = \frac{1}{2} \times 100° = 50°$

Solution 6: (a) 8 cm

Join OC. Then OC = radius = 17 cm



CL = ½ CD = ½ × 30 =15 cm

In right ΔOLC, we have:

$OL^2 = OC^2 - CL^2$

$\Rightarrow OL^2 = (17)^2 - (15)^2$

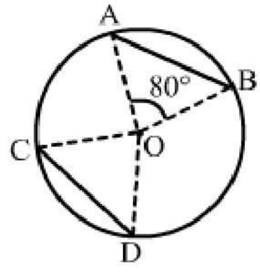
$\Rightarrow OL^2 = (289 - 225)$

$\Rightarrow OL^2 = 64$

$\Rightarrow OL = 8cm$

∴ Distance of CD from AB = 8 cm

Solution 7: (b) 80°

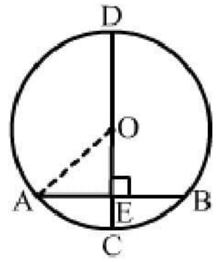


Given: AB = CD

We know that equal chords of a circle subtend equal angles at the centre.

$\therefore \angle COD = \angle AOB = 80°$

Solution 8: (c) 7.5 cm



Let OA = OC = r cm.

Then OE = (r – 3) cm and

AE = ½ AB

= ½ × 12

=6 cm

Now, in right ΔOAE, we have:

$OA^2 = OE^2 + AE^2$

$\Rightarrow (r)^2 = (r-3)^2 + 6^2$

$\Rightarrow r^2 = r^2 + 9 - 6r + 36$

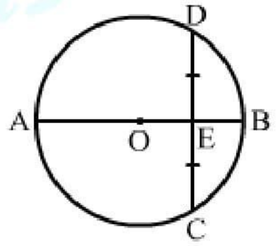
$\Rightarrow 6r = 45$

$\Rightarrow r = \frac{45}{6}$

$\Rightarrow r = 7.5$ cm

Hence, the required radius of the circle is 7.5 cm.

Solution 9: (a) 10 cm



Let the radius of the circle be r cm.

Let OD = OB = r cm.

Then OE = (r – 4) cm and ED = 8 cm

Now, in right ΔOED, we have:

$OD^2 = OE^2 + ED^2$

$\Rightarrow (r)^2 = (r-4)^2 + 8^2$

$\Rightarrow r^2 = r^2 + 16 - 8r + 64$

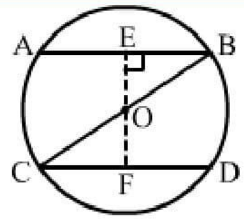
$\Rightarrow 8r = 80$

$\Rightarrow r = 10$ cm

Hence, the required radius of the circle is 10 cm.

Solution 10: (d) 10 cm

Draw $OE \perp AB$ and $OF \perp CD$.



In Δ OEB and ΔOFC, we have:

$OB = OC$ (Radius of a circle)

$\angle BOE = \angle COF$ (Vertically opposite angles)

$\angle OEB = \angle OFC$ (90° each)

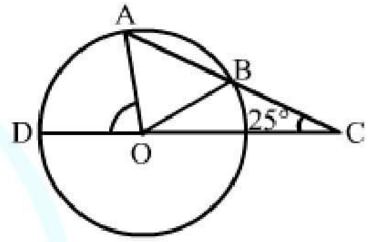
$\therefore \Delta OEB \cong \Delta OFC$ (By AAS congruency rule)

$\therefore OE = OF$

Chords equidistant from the centre are equal.

$\therefore CD = AB = 10$ cm

Solution 11: (b) 75°



$OB = BC$ (Given)

$\Rightarrow \angle BOC = \angle BCO = 25°$

Exterior $\angle OBA = \angle BOC + \angle BCO = (25° + 25°) = 50°$

$OA = OB$ (Radius of a circle)

$\Rightarrow \angle OAB = \angle OBA = 50°$

In Δ AOC, side CO has been produced to D.

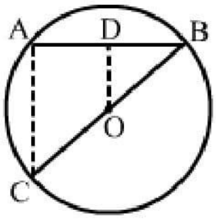
$\therefore$ Exterior $\angle AOD = \angle OAC + \angle ACO$

$= \angle OAB + \angle BCO$

$= (50° + 25°)$

$= 75°$

Solution12: (b) 12 cm



$OD \perp AB$

i.e., D is the mid point of AB.

Also, O is the mid point of BC.

Now, in Δ BAC, D is the mid point of AB and O is the mid point of BC.

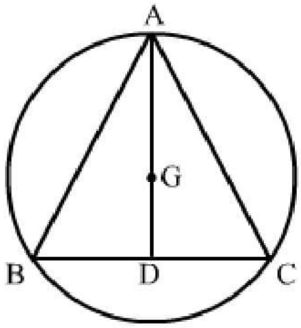
$\therefore OD = \frac{1}{2}AC$ (By mid point theorem)

$\Rightarrow AC = 2OD = (2 \times 6)$ cm $= 12$ cm

Solution 13: (c) $3\sqrt{3}$ cm

Let ΔABC be an equilateral triangle of side 9 cm.

Let AD be one of its medians.



Then $AD \perp BC$ and $BD = 4.5$ cm

$\therefore AD = \sqrt{AB^2 - BD^2}$ [by pythagoras theorem]

$$= \sqrt{9^2 - \left(\frac{9}{2}\right)^2}$$

$$= \sqrt{81 - \frac{9}{4}}$$

$$= \sqrt{\frac{324 - 9}{4}}$$

$$= \sqrt{\frac{243}{4}}$$

$$= \sqrt{\frac{81 \times 3}{4}}$$

$$= \frac{9\sqrt{3}}{2}$$

Let G be the centroid of ΔABC.

Then AG : GD = 2 : 1

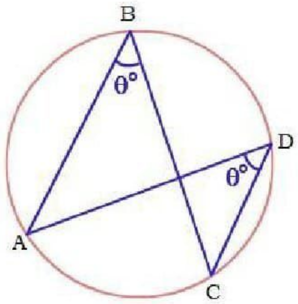
$\therefore$ Radius $= AG = \frac{2}{3}AD = \left(\frac{2}{3} \times \frac{9\sqrt{3}}{2}\right)$ cm $= 3\sqrt{3}$ cm

Solution 14: (c) 90°

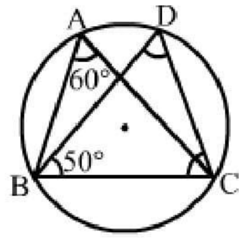
The angle in a semicircle measures 90°.

Solution 15: (a) equal

The angles in the same segment of a circle are equal.



Solution 16: (c) 70°



$\angle BDC = \angle BAC = 60°$ (Angles in the same segment of a circle)

In Δ BDC, we have:

$\angle DBC + \angle BDC + \angle BCD = 180°$ (Angle sum property of a triangle)

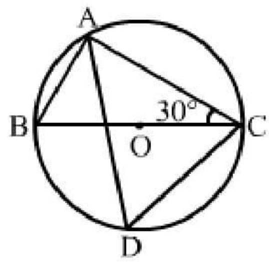
$\therefore 50° + 60° + \angle BCD = 180°$

$\Rightarrow \angle BCD = 180° - (50° + 60°)$

$= (180° - 110°)$

$= 70°$

Solution 17: (c) 60°



Angles in a semi circle measure 90°.

$\therefore \angle BAC = 90°$

In Δ ABC, we have:

$\angle BAC + \angle ABC + \angle BCA = 180°$ (Angle sum property of a triangle)

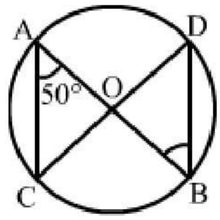
$\therefore 90° + \angle ABC + 30° = 180°$

$\Rightarrow \angle ABC = (180° - 120°)$

$\Rightarrow \angle ABC = 60°$

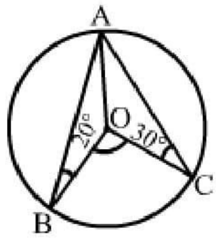
$\therefore \angle CDA = \angle ABC = 60°$ (Angles in the same segment of a circle)

Solution 18: (b) 50°



$\angle ODB = \angle OAC = 50°$ (Angles in the same segment of a circle)

Solution 19: (c) 100°



In Δ OAB, we have:

OA = OB (Radii of a circle)

$\Rightarrow \angle OAB = \angle OBA = 20°$

In ΔOAC, we have:

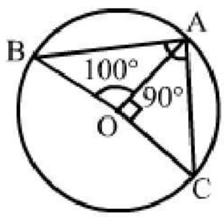
OA = OC (Radii of a circle)

$\Rightarrow \angle OAC = \angle OCA = 30°$

Now, $\angle BAC = (20° + 30°) = 50°$

$\therefore \angle BOC = (2 \times \angle BAC) = (2 \times 50°) = 100°$

Solution 20: (a) 85°



We have:

$\angle BOC + \angle BOA + \angle AOC = 360°$

$\Rightarrow \angle BOC + 100° + 90° = 360°$

$\Rightarrow \angle BOC = (360° - 190°)$

$\Rightarrow \angle BOC = 170°$

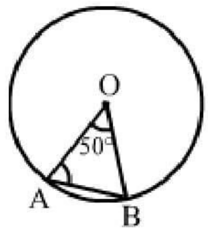
$\therefore \angle BAC = ½ \times \angle BOC$

$= ½ \times 170°$

$= 85°$

Solution 21: (d) 65°

We have:



OA = OB (Radii of a circle)

Let $\angle OAB = \angle OBA = x°$

In ΔOAB, we have:

$x° + x° + 50° = 180°$ (Angle sum property of a triangle)

$\Rightarrow 2x° = (180° - 50°)$

$\Rightarrow 2x° = 130°$

$\Rightarrow x = 130°/2$

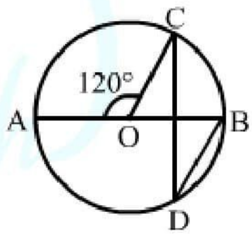
$\Rightarrow x = 65°$

Hence, $\angle OAB = 65°$

Solution 22: (c) 30°

$\angle COB = 180° - 120° = 60°$ (Linear pair)

Now, arc BC subtends $\angle COB$ at the centre and $\angle BDC$ at the point D of the remaining part of the circle.



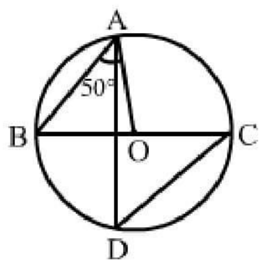
$\therefore \angle COB = 2\angle BDC$

$\Rightarrow \angle BDC = ½ \angle COB$

$= ½ \times 60°$

$= 30°$

Solution 23: (b) 50°



We have:

OA = OB (Radii of a circle)

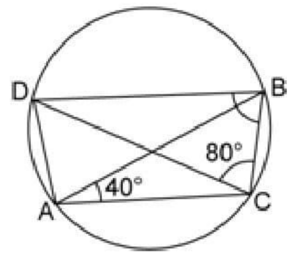
$\Rightarrow \angle OBA = \angle OAB = 50°$

$\therefore \angle CDA = \angle OBA = 50°$ (Angles in the same segment of a circle)

Solution 24: (b) 60°

We have:

$\angle CDB = \angle CAB = 40°$ (Angles in the same segment of a circle)



In Δ CBD, we have:

$\angle CDB + \angle BCD + \angle CBD = 180°$ (Angle sum property of a triangle)

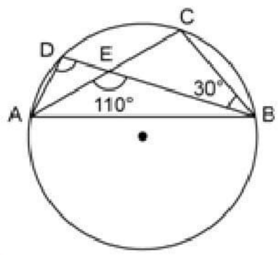
$\Rightarrow 40° + 80° + \angle CBD = 180°$

$\Rightarrow \angle CBD = (180° - 120°)$

$\Rightarrow \angle CBD = 60°$

Solution 25: (c) 80°

We have:



$\angle AEB + \angle CEB = 180°$ (Linear pair angles)

$\Rightarrow 110° + \angle CEB = 180°$

$\Rightarrow \angle CEB = (180° - 110°)$

$\Rightarrow \angle CEB = 70°$

In ΔCEB, we have:

$\angle CEB + \angle EBC + \angle ECB = 180°$ (Angle sum property of a triangle)

$\Rightarrow 70° + 30° + \angle ECB = 180°$

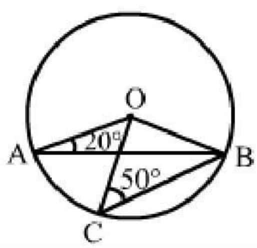
$\Rightarrow \angle ECB = (180° - 100°)$

$\Rightarrow \angle ECB = 80°$

The angles in the same segment are equal.

Thus, $\angle ADB = \angle ECB = 80°$

Solution 26: (d) 60°



We have:

OA = OB (Radii of a circle)

$\Rightarrow \angle OBA = \angle OAB = 20°$

In ΔOAB, we have:

$\angle OAB + \angle OBA + \angle AOB = 180°$ (Angle sum property of a triangle)

$\Rightarrow 20° + 20° + \angle AOB = 180°$

$\Rightarrow \angle AOB = (180° - 40°)$

$\Rightarrow \angle AOB = 140°$

Again, we have:

$OB = OC$ (Radii of a circle)

$\Rightarrow \angle OBC = \angle OCB = 50°$

In ΔOCB, we have:

$\angle OCB + \angle OBC + \angle COB = 180°$ (Angle sum property of a triangle)

$\Rightarrow 50° + 50° + \angle COB = 180°$

$\Rightarrow \angle COB = (180° - 100°)$

$\Rightarrow \angle COB = 80°$

Since $\angle AOB = 140°$, we have:

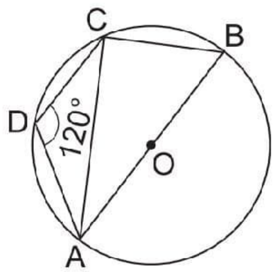
$\angle AOC + \angle COB = 140°$

$\Rightarrow \angle AOC + 80° = 140°$

$\Rightarrow \angle AOC = (140° - 80°)$

$\Rightarrow \angle AOC = 60°$

Solution 27: (b) 30°



We have:

$\angle ABC + \angle ADC = 180°$ (Opposite angles of a cyclic quadrilateral)

$\Rightarrow \angle ABC + 120° = 180°$

$\Rightarrow \angle ABC = (180° - 120°)$

$\Rightarrow \angle ABC = 60°$

Also, $\angle ACB = 90°$ (Angle in a semicircle)

In ΔABC, we have:

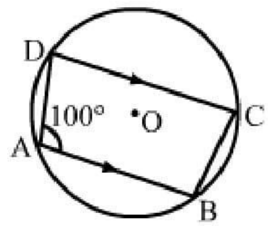
$\angle BAC + \angle ACB + \angle ABC = 180°$ (Angle sum property of a triangle)

$\Rightarrow \angle BAC + 90° + 60° = 180°$

$\Rightarrow \angle BAC = (180° - 150°)$

$\Rightarrow \angle BAC = 30°$

Solution 28: (b) 100°



Since ABCD is a cyclic quadrilateral, we have:

$\angle BAD + \angle BCD = 180°$ (Opposite angles of a cyclic quadrilateral)

$\Rightarrow 100° + \angle BCD = 180°$

$\Rightarrow \angle BCD = (180° - 100°)$

$\Rightarrow \angle BCD = 80°$

Now, AB || DC and CB is the transversal.

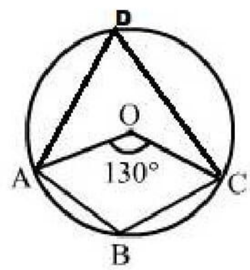
$\therefore \angle ABC + \angle BCD = 180°$

$\Rightarrow \angle ABC + 80° = 180°$

$\Rightarrow \angle ABC = (180° - 80°)$

$\Rightarrow \angle ABC = 100°$

Solution 29: (c) 115°



Take a point D on the remaining part of the circumference.

Join AD and CD.

Then $\angle ADC = \frac{1}{2} \angle AOC = \frac{1}{2} \times 130° = 65°$

In cyclic quadrilateral ABCD, we have:

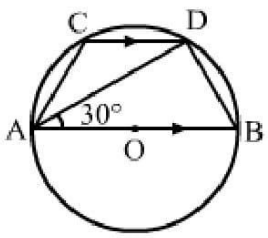
$\angle ABC + \angle ADC = 180°$ (Opposite angles of a cyclic quadrilateral)

$\Rightarrow \angle ABC + 65° = 180°$

$\Rightarrow \angle ABC = (180° - 65°)$

$\Rightarrow \angle ABC = 115°$

Solution 30: (a) 30°



$\angle ADC = \angle BAD = 30°$ (Alternate angles)

$\angle ADB = 90°$ (Angle in semicircle)

$\therefore \angle CDB = \angle ADB + \angle ADC$

$= (90° + 30°)$

$= 120°$

But ABCD being a cyclic quadrilateral, we have:

$\angle BAC + \angle CDB = 180°$

$\Rightarrow \angle BAD + \angle CAD + \angle CDB = 180°$

$\Rightarrow 30° + \angle CAD + 120° = 180°$

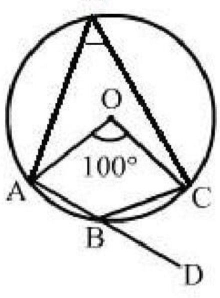
$\Rightarrow \angle CAD = (180° - 150°)$

$\Rightarrow \angle CAD = 30°$

Solution 31: (a) 50°

Take a point E on the remaining part of the circumference.

Join AE and CE.

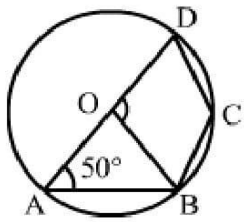


Then ∠AEC = ½ ∠AOC = ½ × 100° = 50°

Now, side AB of the cyclic quadrilateral ABCE has been produced to D.

∴ Exterior ∠CBD = ∠AEC = 50°

Solution 32: (c) 100°



OA = OB (Radii of a circle)

⇒ ∠OBA = ∠OAB = 50°

In Δ OAB, we have:

∠ OAB + ∠OBA + ∠AOB = 180° (Angle sum property of a triangle)

⇒ 50° + 50° + ∠AOB = 180°

⇒ ∠AOB = (180° – 100°)

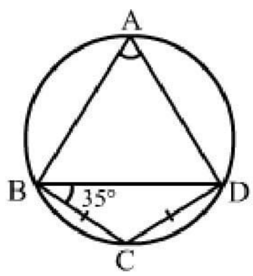
⇒ ∠AOB = 80°

Since ∠AOB + ∠BOD = 180° (Linear pair)

∴ ∠BOD = (180° – 80°)

= 100°

Solution 33: (b) 70°



BC = CD (given)

⇒ ∠BDC = ∠CBD = 35°

In Δ BCD, we have:

∠BCD + BDC + ∠CBD = 180° (Angle sum property of a triangle)

⇒ ∠BCD + 35° + 35° = 180°

⇒ ∠BCD = (180° – 70°)

⇒ ∠BCD = 110°

In cyclic quadrilateral ABCD, we have:

$\angle BAD + \angle BCD = 180°$

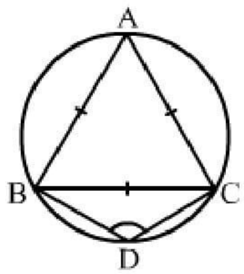
$\Rightarrow \angle BAD + 110° = 180°$

$\therefore \angle BAD = (180° - 110°)$

$= 70°$

Solution 34: (c) 120°

Since ΔABC is an equilateral triangle, each of its angle is 60°.



$\therefore \angle BAC = 60°$

In a cyclic quadrilateral ABCD, we have:

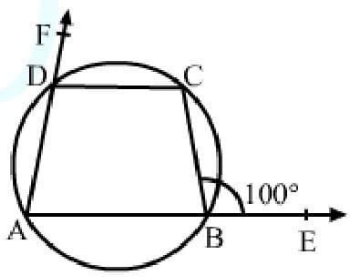
$\angle BAC + \angle BDC = 180°$

$\Rightarrow 60° + \angle BDC = 180°$

$\Rightarrow \angle BDC = (180° - 60°)$

$\Rightarrow \angle BDC = 120°$

Solution 35: (b) 80°



In a cyclic quadrilateral ABCD, we have:

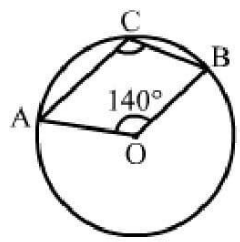
Interior opposite angle, $\angle ADC$ = exterior $\angle CBE = 100°$

$\therefore \angle CDF = (180° - \angle ADC)$

$= (180° - 100°)$

$= 80°$

Solution 36: (c) 110°



Join AB.

Then chord AB subtends $\angle AOB$ at the centre and $\angle ADB$ at a point D of the remaining parts of a circle.

$\therefore \angle AOB = 2\angle ADB$

$\Rightarrow \angle ADB = ½ \angle AOB$

$= ½ \times 140°$

$= 70°$

In the cyclic quadrilateral, we have:

$\angle ADB + \angle ACB = 180°$

$\Rightarrow 70° + \angle ACB = 180°$

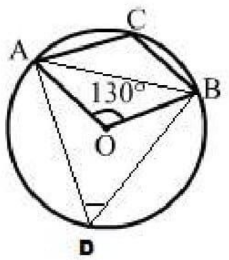
$\therefore \angle ACB = (180° – 70°)$

$= 110°$

Solution 37: (c) 115°

Join AB.

Then chord AB subtends ∠AOB at the centre and ∠ADB at a point D of the remaining parts of a circle.



$\therefore \angle AOB = 2\angle ADB$

$\Rightarrow \angle ADB = ½ \angle AOB$

$= ½ \times 130°$

$=65°$

In cyclic quadrilateral, we have:

$\angle ADB + \angle ACB = 180°$

$\Rightarrow 65° + \angle ACB = 180°$

$\therefore \angle ACB = (180°– 65°) = 115°$

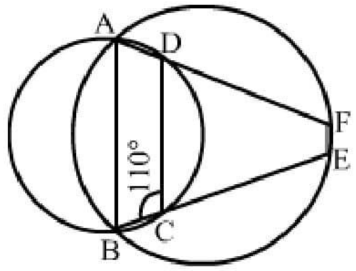
Solution 38: (d) 110°

Since ABCD is a cyclic quadrilateral, we have:

$\angle BAD + \angle BCD = 180°$

$\Rightarrow \angle BAD + 110° = 180°$

$\Rightarrow \angle BAD = (180° - 110°) = 70°$



Similarly, in ABEF, we have:

$\angle BAD + \angle BEF = 180°$

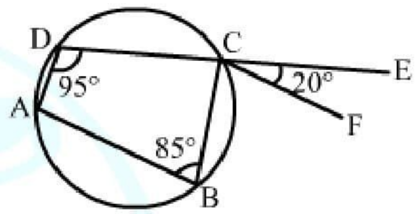
$\Rightarrow 70° + \angle BEF = 180°$

$\Rightarrow \angle BEF = (180° - 70°)$

$\Rightarrow \angle BEF = 110°$

Solution 39: (c) 105°

We have:



$\angle ABC + \angle ADC = 180°$

$\Rightarrow \angle ABC + 95° = 180°$

$\Rightarrow \angle ABC = (180° - 95°)$

$\Rightarrow \angle ABC = 85°$

Now, CF || AB and CB is the transversal.

$\therefore \angle BCF = \angle ABC = 85°$ (Alternate interior angles)

$\Rightarrow \angle BCE = (85° + 20°)$

$\Rightarrow \angle BCE = 105°$

Now,

$\angle DCB = 180° - \angle BCE$

$= (180° - 105°)$

$= 75°$

Now, $\angle BAD + \angle BCD = 180°$

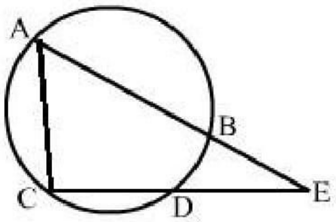
$\Rightarrow \angle BAD + 75° = 180°$

$\Rightarrow \angle BAD = (180° - 75°)$

$\Rightarrow \angle BAD = 105°$

Solution 40: (c) 8.5 cm

Join AC.



Then AE : CE = DE : BE (Intersecting secant theorem)

∴ AE × BE = DE × CE

Let CD = x cm

Then AE = (AB + BE) = (11 + 3) cm = 14 cm;

BE = 3cm;

CE = (x + 3.5) cm; DE = 3.5 cm

∴ 14 × 3 = (x + 3.5) × 3.5

$\Rightarrow x + 3.5 = 14 \times 33.5$

$\Rightarrow x + 3.5 = 423.5$

$\Rightarrow x + 3.5 = 12$

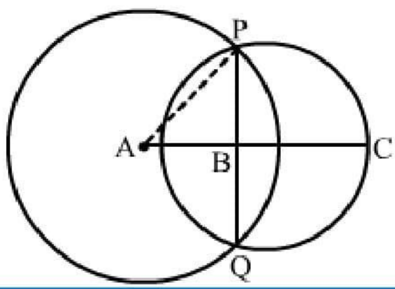
$\Rightarrow x = (12 - 3.5)$

$\Rightarrow x = \ 8.5\ cm$

Hence, CD = 8.5 cm

Solution 41: (b) 6 cm

We know that the line joining their centres is the perpendicular bisector of the common chord.



Join AP.

Then AP = 5 cm; AB = 4 cm

Also, $AP^2 = BP^2 + AB^2$

$\Rightarrow BP^2 = AP^2 - AB^2$

$\Rightarrow BP^2 = 5^2 - 4^2$

$\Rightarrow BP = 3$ cm

$\therefore \Delta ABP$ is a right angled and $PQ = 2 \times BP = (2 \times 3)$ cm $= 6$ cm

Solution 42: (c) 60°

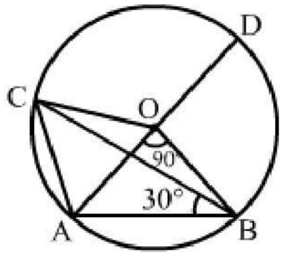
We have:

$\angle AOB = 2\angle ACB$

$\Rightarrow \angle ACB = \frac{1}{2} \times \angle AOB$

$\Rightarrow \angle ACB = \frac{1}{2} \times 90°$

$\Rightarrow \angle ACB = 45°$



$\angle ACB = \frac{1}{2} \angle AOB$

$\Rightarrow \angle ACB = \frac{1}{2} \times 90°$

$\Rightarrow \angle ACB = 45°$

$\angle COA = 2\angle CBA$

$\Rightarrow \angle COA = (2 \times 30°)$

$\Rightarrow \angle COA = 60°$

$\therefore \angle COD = 180° - \angle COA$

$\Rightarrow \angle COD = (180° - 60°)$

$\Rightarrow \angle COD = 120°$

Now, $\angle CAO = \frac{1}{2} \angle COD$

$= \frac{1}{2} \times 120°$

$= 60°$